```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py
然后在浏览器打开 http://127.0.0.1:5000
"""
from app import create_app
from app.models import db
import sys
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")    # 控制台中文正常显示
app = create_app()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()    # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)    # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("  历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0")
    print("=" * 58)
    print("  访问地址：http://127.0.0.1:5000")
    print("  演示账号：admin    密码：admin123")
    print("  停止运行：按 Ctrl+C")
    print("=" * 58)
    app.run(host="127.0.0.1", port=5000, debug=True)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report
```

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app.register_blueprint(auth.bp)
app.register_blueprint(project.bp)
app.register_blueprint(unit.bp)
app.register_blueprint(standard.bp)
app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
# 全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for
from flask_login import current_user, login_user
PUBLIC_ENDPOINTS = {"auth.login", "static"}
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航项：集中定义，模板只负责渲染；active 用于高亮当前页
@app.context_processor
def inject_nav():
    path = request.path or "/"
    items = [
        ("首页", "/", path == "/"),
        ("项目与档案", "/project/", path.startswith("/project
")),
        ("限制性判定", "/constraint/",
path.startswith("/constraint")),
        ("标准知识库", "/standard/", path.startswith("/standa
rd")),
        ("AHP 权重", "/ahp/", path.startswith("/ahp")),
        ("指标数据", "/indicator/", path.startswith("/indica
tor")),
        ("适宜性评价", "/evaluate/", path.startswith("/evalua
te")),
        ("报告输出", "/report/", path.startswith("/report"))
,
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            ("用户与日志", "/users", path.startswith(("/users",
    "/logs"))),
        ]
        return {"nav_items": items}
    # 操作审计：统一在响应阶段记录成功的写操作（M1 操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（M2 遥感影像挂接、M7 批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
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    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（M1 用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志（M1）。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（M2 项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
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    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑（M2）。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接（M2）
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统（M2）：标准化基准值，与图斑绑定。"""
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    __tablename__ = "reference_ecosystems"                          251
    id = db.Column(db.Integer, primary_key=True)                    252
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat        253
    ches.id"), nullable=False)                                      254
    coverage_benchmark = db.Column(db.Float, default=75.0)          255
    # 植被覆盖度基准 %                                              256
    shannon_benchmark = db.Column(db.Float, default=0.7)            257
    # Shannon-Wiener 基准                                           258
    dominant_species = db.Column(db.String(256),                    259
    nullable=True)  # 建群种                                       260
    soil_benchmark = db.Column(db.String(256),                      261
    nullable=True)     # 土壤基准                                   262
class Standard(db.Model):                                           263
    """地方标准 / 政策文件（M3 政策与标准知识库）。"""                 264
    __tablename__ = "standards"                                     265
    id = db.Column(db.Integer, primary_key=True)                    266
    province = db.Column(db.String(32), nullable=False)             267
    # 省份                                                          268
    name = db.Column(db.String(256), nullable=False)                269
    # 名称                                                          270
    doc_number = db.Column(db.String(128), nullable=True)           271
    # 标准编号 / 文号                                               272
    publish_date = db.Column(db.String(32), nullable=True)          273
    implement_date = db.Column(db.String(32), nullable=True)        274
    # 类别：地方标准 / 政策文件                                     275
    category = db.Column(db.String(32), nullable=False,             276
    default="地方标准")                                             277
    summary = db.Column(db.Text, nullable=True)                     278
    # 特点概述                                                      279
    rules = db.relationship("StandardRule",                         280
    backref="standard",                                             281
                                cascade="all, delete-orphan")       282
class StandardRule(db.Model):                                       283
    """结构化规则（M3）：将标准条文拆解为可执行规则。"""             284
    __tablename__ = "standard_rules"                                285
    id = db.Column(db.Integer, primary_key=True)                    286
    standard_id = db.Column(db.Integer,                             287
    db.ForeignKey("standards.id"), nullable=False)                  288
    item = db.Column(db.String(128), nullable=False)                289
    # 判定项                                                        290
    operator = db.Column(db.String(8), nullable=False)              291
    # 比较符：> >= < <= ==                                          292
    threshold = db.Column(db.String(64), nullable=True)             293
    # 阈值                                                          294
    condition_type = db.Column(db.String(32),                       295
    default="必要条件")  # 必要条件 / 充分条件                      296
    region = db.Column(db.String(64), nullable=True)                297
    # 适用分区（宁夏分区差异化）                                    298
    note = db.Column(db.Text, nullable=True)                        299
class ConstraintCheck(db.Model):                                    300
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    """限制性判定记录（M4）。"""                                                301
    __tablename__ = "constraint_checks"                                         302
    id = db.Column(db.Integer, primary_key=True)                                303
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat                    304
    ches.id"), nullable=False)                                                  305
    policy_pass = db.Column(db.Boolean, nullable=True)                          306
    # 政策约束                                                                  307
    location_pass = db.Column(db.Boolean, nullable=True)                        308
    # 区位约束                                                                  309
    pollution_pass = db.Column(db.Boolean, nullable=True)                       310
    # 土壤污染                                                                  311
    geo_pass = db.Column(db.Boolean, nullable=True)                             312
    # 地质灾害                                                                  313
    farmland_pass = db.Column(db.Boolean, nullable=True)                        314
    # 耕地属性                                                                  315
    passed = db.Column(db.Boolean, nullable=True)                               316
    # 总体是否通过                                                              317
    reason = db.Column(db.Text, nullable=True)                                  318
    # 不通过原因                                                                319
    checked_at = db.Column(db.DateTime,                                         320
    default=datetime.utcnow)                                                    321
class MicroUnit(db.Model):                                                      322
    """微地质单元（M5）：岩质边坡 / 碎石边坡 / 平台。"""                        323
    __tablename__ = "micro_units"                                               324
    id = db.Column(db.Integer, primary_key=True)                                325
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat                    326
    ches.id"), nullable=False)                                                  327
    unit_type = db.Column(db.String(32), nullable=False)                        328
    # 单元类型                                                                  329
    area = db.Column(db.Float, nullable=True)                                   330
    # 面积                                                                      331
    ratio = db.Column(db.Float, nullable=True)                                  332
    # 面积占比（自动计算）                                                      333
    description = db.Column(db.Text, nullable=True)                             334
    # 现场描述                                                                  335
class Indicator(db.Model):                                                      336
    """指标定义（M6）：指标树，含角色与启用单元类型。"""                        337
    __tablename__ = "indicators"                                                338
    id = db.Column(db.Integer, primary_key=True)                                339
    code = db.Column(db.String(32), unique=True,                                340
    nullable=False)  # 指标编码                                                 341
    name = db.Column(db.String(64), nullable=False)                             342
    # 指标名称                                                                  343
    level = db.Column(db.Integer, nullable=False)                               344
    # 1/2/3 级                                                                  345
    parent_code = db.Column(db.String(32), nullable=True)                       346
    # 父指标编码                                                                347
    # 角色：土壤污染指标同时作为限制性指标与评价指标                            348
    is_constraint = db.Column(db.Boolean, default=False)                        349
    is_evaluative = db.Column(db.Boolean, default=True)                         350
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    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案（M6）：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细（M6）：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵（M6）：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
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    level = db.Column(db.Integer, nullable=False,                    401
    default=1)          # 矩阵对应指标层级                            402
    matrix_json = db.Column(db.Text, nullable=False)                 403
    # 判断矩阵（JSON 二维数组）                                       404
    cr = db.Column(db.Float, nullable=True)                          405
    # 该专家矩阵的一致性比例                                          406
    created_at = db.Column(db.DateTime,                              407
    default=datetime.utcnow)                                         408
class IndicatorData(db.Model):                                       409
    """指标实测数据（M7）：按微地质单元组织。"""                     410
    __tablename__ = "indicator_data"                                 411
    id = db.Column(db.Integer, primary_key=True)                     412
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni         413
    ts.id"), nullable=False, unique=True)                            414
    soil_ratio = db.Column(db.Float, nullable=True)                  415
    # 土石比                                                         416
    thickness = db.Column(db.Float, nullable=True)                   417
    # 土层厚度 cm                                                    418
    ph = db.Column(db.Float, nullable=True)                          419
    organic_matter = db.Column(db.Float, nullable=True)              420
    # 有机质 g/kg                                                    421
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null 422
    heavy_metals = db.Column(db.Text, nullable=True)                 423
    coverage = db.Column(db.Float, nullable=True)                    424
    # 植被覆盖度 %                                                   425
    shannon = db.Column(db.Float, nullable=True)                     426
    # Shannon-Wiener 多样性指数                                      427
    slope = db.Column(db.Float, nullable=True)                       428
    # 坡度 °                                                         429
    aspect = db.Column(db.String(16), nullable=True)                 430
    # 坡向：阳坡/阴阳坡/阴坡                                         431
    def heavy_metals_dict(self):                                     432
        return json.loads(self.heavy_metals) if                      433
    self.heavy_metals else {}                                        434
class EvaluationResult(db.Model):                                    435
    """评价结果（M8）。"""                                           436
    __tablename__ = "evaluation_results"                             437
    id = db.Column(db.Integer, primary_key=True)                     438
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat         439
    ches.id"), nullable=False)                                       440
    final_score = db.Column(db.Float, nullable=True)                 441
    # 最终评价因子 F                                                 442
    grade = db.Column(db.String(32), nullable=True)                  443
    # 适宜性等级                                                     444
    measures = db.Column(db.Text, nullable=True)                     445
    # 推荐措施                                                       446
    unit_scores = db.Column(db.Text, nullable=True)                  447
    # 各单元得分 JSON                                                448
    created_at = db.Column(db.DateTime,                              449
    default=datetime.utcnow)                                         450
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class ScoreDetail(db.Model):
    """得分明细（M8）：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import DataRequired, Length
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
class AHPSelectForm(FlaskForm):
    """M6：选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """M1：登录表单（占位，由乙完善）。"""
    username = StringField("用户名", validators=[DataRequired()
```

```
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ------------------------------------------------------------------
    ----------
# 一、微地质单元类型
# ------------------------------------------------------------------
    ----------
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),          # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),              # 不设地形指标
}
# ------------------------------------------------------------------
    ----------
# 二、指标标准化理想极值（论文 4.2.2 节）
# ------------------------------------------------------------------
    ----------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 (cm)
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    (g/kg)，全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 (%)
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 (mg/kg)，《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
```

```
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ----------------------------------------------------------------------
    ------------------
# 三、适宜性等级划分（论文表 4）
# ----------------------------------------------------------------------
    ------------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ----------------------------------------------------------------------
    ------------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ----------------------------------------------------------------------
    ------------------
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:
    1.24,
            7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}
# 一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）
CR_THRESHOLD = 0.1
# ----------------------------------------------------------------------
    ------------------
```

```
#  五、默认权重方案（论文表  3）
#  ----------------------------------------------------------------------
    ------------------
#  论文明确："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，
#  分别计算评价指标权重"。权重由  １０  位专家打分后经  ＡＨＰ  方法确定。
#  下表为论文表  3  所载的归一化权重，作为系统默认方案。
ＤＥＦＡＵＬＴ＿ＷＥＩＧＨＴＳ  =  {
    "岩质边坡":  {
        "ｌ１":  {"植被指标":  ０．４５，  "地形指标":  ０．５５},
        "ｌ２":  {
            "植被指标":  {"植被覆盖度":  ０．６７，  "生物多样性":  ０．３３},
            "地形指标":  {"坡度":  ０．６４，  "坡向":  ０．３６},
        },
    },
    "碎石边坡":  {
        "ｌ１":  {"土壤指标":  ０．４４，  "植被指标":  ０．３１，  "地形指标":  ０．２５},
        "ｌ２":  {
            "土壤指标":  {"土石比":  ０．２５，  "土层厚度":  ０．２５，  "ｐＨ":  ０．１７，
                          "有机质":  ０．１７，  "污染指标":  ０．１７},
            "植被指标":  {"植被覆盖度":  ０．６４，  "生物多样性":  ０．３６},
            "地形指标":  {"坡度":  ０．６７，  "坡向":  ０．３３},
        },
    },
    "平台":  {
        "ｌ１":  {"土壤指标":  ０．５４，  "植被指标":  ０．４６},
        "ｌ２":  {
            "土壤指标":  {"土石比":  ０．２５，  "土层厚度":  ０．２５，  "ｐＨ":  ０．１７，
                          "有机质":  ０．１７，  "污染指标":  ０．１７},
            "植被指标":  {"植被覆盖度":  ０．６１，  "生物多样性":  ０．３９},
        },
    },
}
#  ----------------------------------------------------------------------
    ------------------
#  六、限制性指标（论文表  2  +  ４．１  节流程）
#  ----------------------------------------------------------------------
    ------------------
#  论文摘要与  ４．１  节：四项限制性指标；流程段另明确"是否为耕地"为约束条件
ＣＯＮＳＴＲＡＩＮＴ＿ＩＴＥＭＳ  =  （"政策"，  "区位约束"，  "土壤污染"，  "地质灾害"，  "耕地属性"）
#  ====  ａｐｐ/ｓｅｒｖｉｃｅｓ/ａｈｐ＿ｓｅｒｖｉｃｅ．ｐｙ  ====
#  -*-  ｃｏｄｉｎｇ:  ｕｔｆ-８  -*-
"""ＡＨＰ  权重引擎（Ｍ６）。
严格按论文  ４．２．１  节实现：
-  式（２）（３）（４）：特征根法求权重（行元素乘积  →  ｎ  次方根  →  归一化）
-  式（５）（６）（７）：一致性检验（λｍａｘ、ＣＩ、ＣＲ）
-  论文要求  ＣＲ  <  ０．１；ＲＩ  按阶数查表
"""
ｉｍｐｏｒｔ  ｎｕｍｐｙ  ａｓ  ｎｐ
ｆｒｏｍ  ａｐｐ.ｕｔｉｌｓ.ｃｏｎｓｔａｎｔｓ  ｉｍｐｏｒｔ  ＲＩ＿ＴＡＢＬＥ，  ＣＲ＿ＴＨＲＥＳＨＯＬＤ
ｄｅｆ  ｍａｔｒｉｘ＿ｆｒｏｍ＿ｕｐｐｅｒ(ｕｐｐｅｒ):
```

```
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict:{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
```

```
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（M8 的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
```

```
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
```

```
    total = 0.0                                                          801
    for metal, extreme in HEAVY_METAL_LIMIT.items():                     802
        value = heavy_metals.get(metal)                                  803
        total += std_cost(value, extreme) if value is not                804
    None else 1.0                                                        805
    return total / len(HEAVY_METAL_LIMIT)                                806
def standardize_unit(unit_type, data, weights=None):                     807
    """对单个微地质单元的全部指标做标准化。                                  808
    参数                                                                 809
    ----                                                                 810
    unit_type : str，岩质边坡 / 碎石边坡 / 平台                            811
    data : dict，实测数据，字段见接口契约：                                 812
        soil_ratio, thickness, ph, organic_matter,                       813
    heavy_metals(dict),                                                  814
        coverage, shannon, slope, aspect                                 815
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG     816
    HTS 结构），                                                          817
            未提供时用默认值。                                            818
    返回                                                                 819
    ----                                                                 820
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }   821
    标准化值可能为 None，表示该指标不适用或方法待定。                       822
    """                                                                  823
    from app.utils.constants import DEFAULT_WEIGHTS                      824
    if weights is None:                                                  825
        weights = DEFAULT_WEIGHTS[unit_type]                             826
    l1_enabled = weights["l1"].keys()                                    827
    l2w = weights["l2"]                                                  828
    result = {}                                                          829
    # ---- 土壤指标 ----                                                 830
    if "土壤指标" in l1_enabled:                                          831
        result["土壤指标"] = {                                            832
            "土石比": (data.get("soil_ratio"),                           833
                       std_benefit(data.get("soil_ratio"),               834
    IDEAL_SOIL_RATIO),                                                   835
                       l2w["土壤指标"]["土石比"]),                        836
            "土层厚度": (data.get("thickness"),                          837
                         std_benefit(data.get("thickness"),              838
    IDEAL_THICKNESS),                                                    839
                         l2w["土壤指标"]["土层厚度"]),                    840
            "pH": (data.get("ph"),                                       841
                   std_mid_ph(data.get("ph"),                            842
    *IDEAL_PH_RANGE),                                                    843
                   l2w["土壤指标"]["pH"]),                               844
            "有机质": (data.get("organic_matter"),                       845
                       std_benefit(data.get("organic_matter")            846
    , IDEAL_ORGANIC),                                                    847
                       l2w["土壤指标"]["有机质"]),                       848
            "污染指标": (data.get("heavy_metals") or {},                 849
                         pollution_score(data.get("heavy_meta            850
```

```
        ls") or {}),                                                    851
                                    l2w["土壤指标"]["污染指标"]),          852
            }                                                           853
        # ---- 植被指标 ----                                           854
        if "植被指标" in l1_enabled:                                    855
            result["植被指标"] = {                                      856
                "植被覆盖度": (data.get("coverage"),                    857
                                    std_benefit(data.get("coverage"),   858
        IDEAL_COVERAGE),                                                859
                                    l2w["植被指标"]["植被覆盖度"]),      860
                "生物多样性": (data.get("shannon"),                     861
                                    std_benefit(data.get("shannon"),    862
        IDEAL_SHANNON),                                                 863
                                    l2w["植被指标"]["生物多样性"]),      864
            }                                                           865
        # ---- 地形指标 ----                                           866
        if "地形指标" in l1_enabled:                                    867
            result["地形指标"] = {                                      868
                "坡度": (data.get("slope"),                             869
                        std_slope(data.get("slope")),                   870
                        l2w["地形指标"]["坡度"]),                       871
                "坡向": (data.get("aspect"),                            872
                        aspect_score(data.get("aspect")),               873
                        l2w["地形指标"]["坡向"]),                       874
            }                                                           875
        return result                                                   876
# ==== app/services/evaluation_service.py ====                        877
# -*- coding: utf-8 -*-                                               878
"""适宜性评价引擎（M8）。                                               879
按论文式(9) 实现：                                                      880
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）          881
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                  882
- 等级判定按论文表 4。                                                  883
"""                                                                     884
from app.utils.constants import DEFAULT_WEIGHTS, grade_of               885
from app.services import standardization as std                         886
def score_unit(unit_type, data, weights=None):                          887
    """计算单个微地质单元的评价因子 F'。                                888
    返回                                                                889
    ----                                                                890
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:       891
    [...] }                                                             892
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，           893
    则按其余项权重归一化后计算，保证单元得分始终有定义。                894
    """                                                                 895
    if weights is None:                                                 896
        weights = DEFAULT_WEIGHTS[unit_type]                            897
    std_result = std.standardize_unit(unit_type, data,                  898
    weights)                                                            899
    l1_weights = weights["l1"]                                          900
```

```
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
raw,
                    "std_value": None, "weight": w2,
"contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    unit_score = sum(l1_weights[k] * l1_scores[k] for k in
l1_scores)
    return {"l1_scores": l1_scores, "unit_score":
unit_score, "details": details}
def score_mine(units):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    total = 0.0
    unit_results = []
    for u in units:
        r = score_unit(u["unit_type"], u["data"])
        weighted = u["ratio"] * r["unit_score"]
        total += weighted
        unit_results.append({
            "unit_type": u["unit_type"],
            "ratio": u["ratio"],
            "unit_score": r["unit_score"],
            "weighted": weighted,
```

```
                "l1_scores": r["l1_scores"],
                "details": r["details"],
            })
    grade_info = grade_of(total)
    return {
        "final_score": total,
        "grade": grade_info["grade"],
        "grade_info": grade_info,
        "unit_results": unit_results,
    }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成（M9）：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
        run.font.name = "Times New Roman"
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")
    return h
def _table(doc, headers, rows, widths=None):
    t = doc.add_table(rows=1, cols=len(headers))
    t.style = "Table Grid"
    hdr = t.rows[0].cells
    for i, htext in enumerate(headers):
```

```
            _set_run(hdr[i].paragraphs[0].add_run(), htext,
    10.5, bold=True)
    for row in rows:
        cells = t.add_row().cells
        for i, val in enumerate(row):
            _set_run(cells[i].paragraphs[0].add_run(),
    str(val), 10.5)
    return t
def build_report(patch, mine_result):
    """生成评价报告 Word 文档，返回 BytesIO。
    参数
    ----
    patch : MinePatch，图斑对象
    mine_result : evaluation_service.score_mine 的返回 dict
    """
    doc = Document()
    # 页边距
    for section in doc.sections:
        section.top_margin = Cm(2.5)
        section.bottom_margin = Cm(2.5)
        section.left_margin = Cm(2.5)
        section.right_margin = Cm(2.5)
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d
    %H:%M:%S"),
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,
    space_after=12)
    # 图斑信息
    _heading(doc, "一、图斑基本信息", 1)
    info = [
        ("图斑名称", patch.name),
        ("所在行政区", patch.admin_region or "—"),
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is
    not None else "—"),
        ("开采矿种", patch.mineral_type or "—"),
        ("开采方式", patch.mining_method or "—"),
    ]
    _table(doc, ["项目", "内容"], info)
    # 评价结论
    _heading(doc, "二、评价结论", 1)
    gi = mine_result["grade_info"]
    concl = [
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),
        ("适宜性等级", mine_result["grade"]),
        ("判定依据", gi["meaning"]),
        ("推荐措施", gi["measure"]),
    ]
    _table(doc, ["项目", "结果"], concl)
```

```
    # 各单元得分
    _heading(doc, "三、各微地质单元得分", 1)
    rows = []
    for u in mine_result["unit_results"]:
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"] * 100),
                     "%.4f" % u["unit_score"], "%.4f" % u["weighted"]))
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"], rows)
    # 完整得分明细
    _heading(doc, "四、得分明细", 1)
    for u in mine_result["unit_results"]:
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"], u["ratio"] * 100),
              size=11, bold=True, space_after=4)
        detail_rows = []
        for d in u["details"]:
            raw = "—" if d["raw"] is None else d["raw"]
            std = "—" if d["std_value"] is None else "%.4f" % d["std_value"]
            w = "%.4f" % d["weight"]
            c = "—" if d["contribution"] is None else "%.4f" % d["contribution"]
            detail_rows.append((d["l1"], d["l2"], raw, std, w, c))
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重", "贡献"],
               detail_rows)
        # 一级指标得分
        l1_row = []
        for k, v in u["l1_scores"].items():
            l1_row.append("%s=%.4f" % (k, v))
        _para(doc, "一级指标得分：" + "，".join(l1_row), size=10.5, space_after=10)
    buf = io.BytesIO()
    doc.save(buf)
    buf.seek(0)
    return buf
# ==== app/views/auth.py ====
# -*- coding: utf-8 -*-
"""M1 用户与权限管理 —— 登录、注销、用户维护、操作日志。"""
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                   flash, abort)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import LoginForm, EmptyForm
```

```
from app.models import db, User, OperationLog                    1101
bp = Blueprint("auth", __name__)                                 1102
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",          1103
    "viewer": "查看者"}                                          1104
def require_admin():                                             1105
    """非管理员直接 403，避免误操作基础数据。"""                  1106
    if current_user.role != "admin":                             1107
        abort(403)                                               1108
@bp.route("/")                                                   1109
def index():                                                     1110
    if current_user.is_authenticated:                            1111
        return render_template("auth/home.html")                 1112
    return redirect(url_for("auth.login"))                       1113
@bp.route("/login", methods=["GET", "POST"])                     1114
def login():                                                     1115
    if current_user.is_authenticated:                            1116
        return redirect(url_for("auth.index"))                   1117
    form = LoginForm()                                           1118
    if form.validate_on_submit():                                1119
        user = User.query.filter_by(username=form.username.d1120
    ata).first()                                                 1121
        if user and check_password_hash(user.password_hash,      1122
    form.password.data):                                         1123
            if not user.active:                                  1124
                flash("该账号已停用，请联系管理员", "error")      1125
                return render_template("auth/login.html",        1126
    form=form)                                                   1127
            login_user(user, remember=form.remember.data)        1128
            flash("登录成功", "ok")                              1129
            return redirect(url_for("auth.index"))               1130
        flash("用户名或密码错误", "error")                       1131
    return render_template("auth/login.html", form=form)         1132
@bp.route("/logout")                                             1133
@login_required                                                  1134
def logout():                                                    1135
    logout_user()                                                1136
    flash("已退出登录", "ok")                                    1137
    return redirect(url_for("auth.login"))                       1138
# ---------------------------- 用户维护 ----------------------------1139
    ----                                                         1140
@bp.route("/users")                                              1141
@login_required                                                  1142
def users():                                                     1143
    all_users = User.query.order_by(User.id).all()               1144
    return render_template("auth/users.html",                    1145
    users=all_users,                                             1146
                                  role_labels=ROLE_LABELS,       1147
    form=EmptyForm())                                            1148
@bp.route("/users/add", methods=["GET", "POST"])                 1149
@login_required                                                  1150
```

```
def add_user():
    require_admin()
    if request.method == "POST":
        username = request.form.get("username", "").strip()
        password = request.form.get("password", "")
        role = request.form.get("role", "viewer")
        if not username or not password:
            flash("用户名和密码不能为空", "error")
            return redirect(url_for("auth.add_user"))
        if User.query.filter_by(username=username).first():
            flash("用户名已存在", "error")
            return redirect(url_for("auth.add_user"))
        db.session.add(User(username=username, role=role,
                            password_hash=generate_password_
    hash(password)))
        db.session.commit()
        flash("用户已添加", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/add_user.html")
@bp.route("/users/<int:user_id>/edit", methods=["GET",
    "POST"])
@login_required
def edit_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if request.method == "POST":
        role = request.form.get("role", user.role)
        active = request.form.get("active") == "1"
        # 保护：不能把最后一个管理员降级或停用，否则无人能管理系统
        if user.role == "admin" and (role != "admin" or not
    active):
            others = User.query.filter(User.role == "admin",
                                       User.id != user.id,

    User.active.is_(True)).count()
            if others == 0:
                flash("系统至少需保留一名启用状态的管理员", "error")
                return redirect(url_for("auth.edit_user",
    user_id=user.id))
        user.role = role
        user.active = active
        new_password = request.form.get("password",
    "").strip()
        if new_password:
            user.password_hash = generate_password_hash(new_
    password)
        db.session.commit()
        flash("用户信息已保存", "ok")
        return redirect(url_for("auth.users"))
    return render_template("auth/edit_user.html", user=user,
```

```
                                role_labels=ROLE_LABELS,
    form=EmptyForm())
@bp.route("/users/<int:user_id>/delete", methods=["POST"])
@login_required
def delete_user(user_id):
    require_admin()
    user = db.get_or_404(User, user_id)
    if user.id == current_user.id:
        flash("不能删除当前登录的账号", "error")
        return redirect(url_for("auth.users"))
    if user.role == "admin":
        others = User.query.filter(User.role == "admin",
                                   User.id != user.id,
                                   User.active.is_(True)).co
    unt()
        if others == 0:
            flash("系统至少需保留一名启用状态的管理员", "error")
            return redirect(url_for("auth.users"))
    db.session.delete(user)
    db.session.commit()
    flash("用户「%s」已删除" % user.username, "ok")
    return redirect(url_for("auth.users"))
# ------------------------------ 操作日志 ------------------------------
    ----
@bp.route("/logs")
@login_required
def logs():
    page = request.args.get("page", 1, type=int)
    per_page = 50
    q = OperationLog.query.order_by(OperationLog.id.desc())
    total = q.count()
    records = q.offset((page - 1) * per_page).limit(per_page
    ).all()
    users_map = {u.id: u.username for u in User.query.all()}
    pages = max(1, (total + per_page - 1) // per_page)
    return render_template("auth/logs.html",
    records=records, page=page,
                           pages=pages, total=total,
    users_map=users_map)
# ==== app/views/project.py ====
# -*- coding: utf-8 -*-
"""M2 项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""
from datetime import datetime
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, current_app)
from flask_login import login_required, current_user
from app.forms import EmptyForm
from app.models import db, Project, MinePatch,
    ReferenceEcosystem
```

```
bp = Blueprint("project", __name__, url_prefix="/project")
@bp.route("/")
@login_required
def index():
    projects = Project.query.order_by(Project.id.desc()).all()
    return render_template("project/index.html",
    projects=projects,
                           form=EmptyForm())
@bp.route("/add", methods=["GET", "POST"])
@login_required
def add():
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        p = Project(
            name=request.form.get("name", "").strip(),
            code=request.form.get("code", "").strip() or None,
            description=request.form.get("description", "").strip() or None,
            created_by=current_user.id,
        )
        if not p.name:
            flash("项目名称不能为空", "error")
        else:
            db.session.add(p)
            db.session.commit()
            flash("项目已创建", "ok")
            return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form, project=None)
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])
@login_required
def edit(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit():
        p.name = request.form.get("name", "").strip()
        p.code = request.form.get("code", "").strip() or None
        p.description = request.form.get("description", "").strip() or None
        p.status = request.form.get("status", "进行中")
        db.session.commit()
        flash("项目已保存", "ok")
        return redirect(url_for("project.index"))
    return render_template("project/edit.html", form=form, project=p)
```

```
@bp.route("/<int:pid>/delete", methods=["POST"])
@login_required
def delete(pid):
    p = db.get_or_404(Project, pid)
    db.session.delete(p)
    db.session.commit()
    flash("项目已删除", "ok")
    return redirect(url_for("project.index"))
# ---------------------------- 矿山图斑 ----------------------------
----
@bp.route("/<int:pid>/patches")
@login_required
def patches(pid):
    p = db.get_or_404(Project, pid)
    return render_template("project/patches.html",
    project=p,
                           form=EmptyForm())
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])
@login_required
def patch_delete(patch_id):
    """删除矿山图斑，及其微地质单元与指标数据。"""
    from app.models import MicroUnit, IndicatorData,
    EvaluationResult, \
        ScoreDetail, ConstraintCheck, ReferenceEcosystem
    patch = db.get_or_404(MinePatch, patch_id)
    pid = patch.project_id
    unit_ids = [u.id for u in MicroUnit.query.filter_by(patc
    h_id=patch.id).all()]
    if unit_ids:
        IndicatorData.query.filter(IndicatorData.unit_id.in_
    (unit_ids)).delete(
            synchronize_session=False)
    for r in EvaluationResult.query.filter_by(patch_id=patch
    .id).all():
        ScoreDetail.query.filter_by(result_id=r.id).delete(
            synchronize_session=False)
    EvaluationResult.query.filter_by(patch_id=patch.id).dele
    te(
        synchronize_session=False)
    ConstraintCheck.query.filter_by(patch_id=patch.id).delet
    e(
        synchronize_session=False)
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
```

```
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
```

```
        patch.area_ha = fv("area_ha", float)                          1401
        patch.mineral_type = fv("mineral_type")                       1402
        patch.mining_method = fv("mining_method")                     1403
        patch.stop_years = fv("stop_years", float)                    1404
        patch.responsible_entity_missing =                            1405
    bool(request.form.get("responsible_entity_missing"))              1406
        # 参照生态系统                                                 1407
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa          1408
    tch.id).first()                                                   1409
        if ref is None:                                               1410
            ref = ReferenceEcosystem(patch_id=patch.id)               1411
            db.session.add(ref)                                       1412
        ref.coverage_benchmark = fv("coverage_benchmark",             1413
    float) or 75.0                                                    1414
        ref.shannon_benchmark = fv("shannon_benchmark",               1415
    float) or 0.7                                                     1416
        ref.dominant_species = fv("dominant_species")                 1417
        patch.image_note = fv("image_note")                           1418
        db.session.commit()                                           1419
        flash("图斑已保存", "ok")                                      1420
        return redirect(url_for("project.patches",                    1421
    pid=patch.project_id))                                            1422
    return render_template("project/patch_edit.html",                 1423
    form=form,                                                        1424
                           project=patch.project,                     1425
    patch=patch,                                                      1426
                           reference=ReferenceEcosystem.quer          1427
    y.filter_by(                                                      1428
                               patch_id=patch.id).first())            1429
# ------------------------------ 遥感影像挂接                        1430
    ------------------------------                                    1431
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])            1432
@login_required                                                       1433
def upload_image(patch_id):                                           1434
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""                    1435
    from app.utils import uploads as up                               1436
    patch = db.get_or_404(MinePatch, patch_id)                        1437
    form = EmptyForm()                                                1438
    if not form.validate_on_submit():                                 1439
        flash("表单校验失败，请重试", "error")                         1440
        return redirect(url_for("project.patch_edit",                 1441
    patch_id=patch.id))                                               1442
    rel, err = up.save_upload(request.files.get("image"),             1443
    "patches",                                                        1444
                              current_app.config["ALLOWED_IM          1445
    AGE_EXT"],                                                        1446
                              current_app.config["UPLOAD_FOL          1447
    DER"])                                                            1448
    if err:                                                           1449
        flash(err, "error")                                           1450
```

```
            return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
        # 替换旧影像时清理原文件，避免残留
        if patch.image_path:
            up.remove_upload(patch.image_path,
        current_app.config["UPLOAD_FOLDER"])
        patch.image_path = rel
        patch.image_uploaded_at = datetime.utcnow()
        note = request.form.get("image_note", "").strip()
        if note:
            patch.image_note = note
        db.session.commit()
        flash("遥感影像已挂接", "ok")
        return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
@bp.route("/patch/<int:patch_id>/image/delete",
        methods=["POST"])
@login_required
def delete_image(patch_id):
        from app.utils import uploads as up
        patch = db.get_or_404(MinePatch, patch_id)
        form = EmptyForm()
        if form.validate_on_submit():
            up.remove_upload(patch.image_path,
        current_app.config["UPLOAD_FOLDER"])
            patch.image_path = None
            patch.image_uploaded_at = None
            db.session.commit()
            flash("遥感影像已移除", "ok")
        return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""M5 微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
        redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
        IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
def recalc_ratio(patch_id):
        """按面积重算该图斑各单元面积占比。"""
        units = MicroUnit.query.filter_by(patch_id=patch_id).all
        ()
        total = sum(u.area or 0.0 for u in units)
        if total > 0:
            for u in units:
                u.ratio = round((u.area or 0.0) / total, 6)
```

```
    return total
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
                      description=request.form.get("descript
    ion", "").strip() or None)
        db.session.add(u)
        db.session.flush()
        recalc_ratio(patch.id)
        db.session.commit()
        flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        u.unit_type = request.form.get("unit_type",
    u.unit_type)
        area = request.form.get("area", "").strip()
        u.area = float(area) if area else None
        u.description = request.form.get("description",
    "").strip() or None
        recalc_ratio(u.patch_id)
```

```
            db.session.commit()                                          1551
            flash("微地质单元已保存", "ok")                                1552
            return redirect(url_for("unit.index",                        1553
        patch_id=u.patch_id))                                            1554
        return render_template("unit/edit.html", unit=u,                 1555
        patch=u.patch,                                                   1556
                                    unit_types=UNIT_TYPES, form=form)    1557
@bp.route("/delete/<int:unit_id>", methods=["POST"])                     1558
@login_required                                                          1559
def delete(unit_id):                                                     1560
        u = db.get_or_404(MicroUnit, unit_id)                            1561
        pid = u.patch_id                                                 1562
        name = u.unit_type                                               1563
        # 一并删除该单元的指标数据，避免残留                                  1564
        IndicatorData.query.filter_by(unit_id=u.id).delete(synch         1565
        ronize_session=False)                                            1566
        db.session.delete(u)                                             1567
        db.session.flush()                                               1568
        recalc_ratio(pid)                                                1569
        db.session.commit()                                              1570
        flash("微地质单元「%s」已删除" % name, "ok")                          1571
        return redirect(url_for("unit.index", patch_id=pid))             1572
# ==== app/views/standard.py ====                                        1573
# -*- coding: utf-8 -*-                                                  1574
"""M3 政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""          1575
from flask import Blueprint, render_template, request,                   1576
        redirect, url_for, flash                                         1577
from flask_login import login_required                                   1578
from app.forms import EmptyForm                                          1579
from app.models import db, Standard, StandardRule                        1580
bp = Blueprint("standard", __name__, url_prefix="/standard")             1581
@bp.route("/")                                                           1582
@login_required                                                          1583
def index():                                                             1584
        standards = Standard.query.order_by(Standard.category,           1585
        Standard.province).all()                                         1586
        local = [s for s in standards if s.category == "地方标准"]        1587
        policy = [s for s in standards if s.category != "地方标准"]       1588
        return render_template("standard/index.html",                    1589
        local=local, policy=policy,                                      1590
                                    form=EmptyForm())                    1591
@bp.route("/add", methods=["GET", "POST"])                               1592
@login_required                                                          1593
def add():                                                               1594
        form = EmptyForm()                                               1595
        if request.method == "POST" and form.validate_on_submit(         1596
        ):                                                               1597
            s = Standard(                                                1598
                province=request.form.get("province",                    1599
        "").strip(),                                                     1600
```

```
            name=request.form.get("name", "").strip(),
            doc_number=request.form.get("doc_number",
    "").strip() or None,
            publish_date=request.form.get("publish_date",
    "").strip() or None,
            implement_date=request.form.get("implement_date"
    , "").strip() or None,
            category=request.form.get("category", "地方标准"),
            summary=request.form.get("summary", "").strip()
    or None,
        )
        if not s.province or not s.name:
            flash("省份与名称不能为空", "error")
        else:
            db.session.add(s)
            db.session.commit()
            flash("标准已添加", "ok")
            return redirect(url_for("standard.index"))
    return render_template("standard/edit.html", form=form,
    standard=None)
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])
@login_required
def edit(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        s.province = request.form.get("province",
    "").strip()
        s.name = request.form.get("name", "").strip()
        s.doc_number = request.form.get("doc_number",
    "").strip() or None
        s.publish_date = request.form.get("publish_date",
    "").strip() or None
        s.implement_date = request.form.get("implement_date"
    , "").strip() or None
        s.category = request.form.get("category", "地方标准")
        s.summary = request.form.get("summary", "").strip()
    or None
        db.session.commit()
        flash("已保存", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    return render_template("standard/edit.html", form=form,
    standard=s)
@bp.route("/<int:sid>/delete", methods=["POST"])
@login_required
def delete(sid):
    s = db.get_or_404(Standard, sid)
    StandardRule.query.filter_by(standard_id=sid).delete()
    db.session.delete(s)
```

```
    db.session.commit()
    flash("已删除", "ok")
    return redirect(url_for("standard.index"))
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])
@login_required
def rules(sid):
    s = db.get_or_404(Standard, sid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        r = StandardRule(
            standard_id=s.id,
            item=request.form.get("item", "").strip(),
            operator=request.form.get("operator", ">="),
            threshold=request.form.get("threshold",
    "").strip() or None,
            condition_type=request.form.get("condition_type"
    , "必要条件"),
            region=request.form.get("region", "").strip()
    or None,
            note=request.form.get("note", "").strip() or
    None,
        )
        if not r.item:
            flash("判定项不能为空", "error")
        else:
            db.session.add(r)
            db.session.commit()
            flash("规则已添加", "ok")
        return redirect(url_for("standard.rules", sid=s.id))
    rs = StandardRule.query.filter_by(standard_id=s.id).all(
    )
    return render_template("standard/rules.html",
    form=form, standard=s, rules=rs)
@bp.route("/rule/<int:rid>/delete", methods=["POST"])
@login_required
def rule_delete(rid):
    r = db.get_or_404(StandardRule, rid)
    sid = r.standard_id
    db.session.delete(r)
    db.session.commit()
    flash("规则已删除", "ok")
    return redirect(url_for("standard.rules", sid=sid))
@bp.route("/compare")
@login_required
def compare():
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""
    rules = StandardRule.query.join(Standard).all()
    groups = {}
    for r in rules:
```

```
            groups.setdefault(r.item, []).append(r)
        return render_template("standard/compare.html",
        groups=groups)
# ==== app/views/constraint.py ====
# -*- coding: utf-8 -*-
"""M4 限制性指标判定 —— 一票否决判定。
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, ConstraintCheck
from app.utils.constants import CONSTRAINT_ITEMS
bp = Blueprint("constraint", __name__,
    url_prefix="/constraint")
# 字段名 -> 中文名
FIELDS = [
    ("policy_pass", "政策约束"),
    ("location_pass", "区位约束"),
    ("pollution_pass", "土壤污染"),
    ("geo_pass", "地质灾害"),
    ("farmland_pass", "耕地属性"),
]
@bp.route("/")
@login_required
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    latest = {}
    for p in patches:
        latest[p.id] = ConstraintCheck.query.filter_by(patch
    _id=p.id) \
            .order_by(ConstraintCheck.id.desc()).first()
    return render_template("constraint/index.html",
    patches=patches, latest=latest)
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
@login_required
def check(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    record = ConstraintCheck.query.filter_by(patch_id=patch_
    id) \
        .order_by(ConstraintCheck.id.desc()).first()
    if request.method == "POST" and form.validate_on_submit(
    ):
        if record is None:
            record = ConstraintCheck(patch_id=patch.id)
            db.session.add(record)
```

```
            for field, _ in FIELDS:                                    1751
                setattr(record, field, request.form.get(field)         1752
        == "1")                                                        1753
            record.passed = all(getattr(record, f) for f, _ in         1754
        FIELDS)                                                        1755
            failed = [name for f, name in FIELDS if not                1756
        getattr(record, f)]                                            1757
            record.reason = ("全部通过" if record.passed               1758
                             else "未通过：" + "、".join(failed))      1759
            db.session.commit()                                        1760
            flash("判定结果已保存", "ok")                              1761
            return redirect(url_for("constraint.check",                1762
        patch_id=patch.id))                                            1763
        # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）  1764
        field_states = [(f, name, None if record is None else          1765
        getattr(record, f))                                            1766
                        for f, name in FIELDS]                         1767
        return render_template("constraint/check.html",                1768
        patch=patch, form=form,                                        1769
                               fields=FIELDS,                          1770
        field_states=field_states,                                     1771
                               record=record)                          1772
# ==== app/views/ahp.py ====                                           1773
# -*- coding: utf-8 -*-                                                1774
"""M6 指标体系与 AHP 权重 —— 路由与页面逻辑。"""                       1775
import json                                                            1776
from flask import Blueprint, render_template, request,                 1777
    redirect, url_for, flash                                           1778
from app.forms import AHPSelectForm, EmptyForm                         1779
from app.models import db, Indicator, ExpertMatrix,                    1780
    WeightScheme                                                       1781
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES            1782
from app.services import ahp_service as ahp                            1783
from app.services import indicator_service as ind_svc                  1784
bp = Blueprint("ahp", __name__, url_prefix="/ahp")                     1785
def indicator_list(unit_type, level, l1=None):                         1786
    """返回 (指标名称列表, 默认权重列表)。                             1787
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。  1788
    """                                                                1789
    weights = DEFAULT_WEIGHTS[unit_type]                               1790
    if level == 1:                                                     1791
        names = list(weights["l1"].keys())                             1792
        vec = [weights["l1"][n] for n in names]                        1793
        return names, vec                                              1794
    # level == 2                                                       1795
    l2 = weights["l2"][l1]                                             1796
    names = list(l2.keys())                                            1797
    vec = [l2[n] for n in names]                                       1798
    return names, vec                                                  1799
def l1_choices(unit_type):                                             1800
```

```
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form)
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form)
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
    level = int(request.values.get("level", 1))
    l1 = request.values.get("l1")
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
    range(n)] for i in range(n)],
            "most_inconsistent": bad,
```

```
            }                                                                   1851
        result = None                                                           1852
        if request.method == "POST" and form.validate_on_submit(               1853
        ):                                                                      1854
            try:                                                                1855
                upper = [float(request.form.get("m_%d_%d" % (i,                 1856
        j), "1"))                                                               1857
                         for i in range(n) for j in range(i +                   1858
        1, n)]                                                                  1859
                result = compute(upper)                                         1860
            except (ValueError, ZeroDivisionError):                             1861
                flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",          1862
        "error")                                                                1863
        elif request.values.get("calc") == "1":                                 1864
            # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性                          1865
            result = compute([default[i][j] for i in range(n)                   1866
        for j in range(i + 1, n)])                                              1867
        return render_template(                                                 1868
            "ahp/matrix.html",                                                  1869
            form=form, unit_type=unit_type, level=level, l1=l1,                 1870
            names=names, vec=vec, default=default,                              1871
        result=result,                                                          1872
        )                                                                       1873
# ------------------------------ 指标维护 ------------------------------1874
    ----                                                                        1875
@bp.route("/indicators")                                                        1876
def indicators():                                                               1877
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                                  1878
    if Indicator.query.count() == 0:                                            1879
        ind_svc.seed_indicators()                                               1880
    all_ind = Indicator.query.order_by(Indicator.level,                         1881
    Indicator.id).all()                                                         1882
    by_parent = {}                                                              1883
    for i in all_ind:                                                           1884
        by_parent.setdefault(i.parent_code, []).append(i)                       1885
    return render_template("ahp/indicators.html",                               1886
    by_parent=by_parent,                                                        1887
                                all_ind=all_ind,                                1888
    unit_types=UNIT_TYPES,                                                      1889
                                type_label=ind_svc.ideal_type_lab1890
    el,                                                                         1891
                                form=EmptyForm())                               1892
@bp.route("/indicators/<code>/edit", methods=["GET",                            1893
    "POST"])                                                                    1894
def indicator_edit(code):                                                       1895
    ind = Indicator.query.filter_by(code=code).first_or_404(                    1896
    )                                                                           1897
    form = EmptyForm()                                                          1898
    if request.method == "POST" and form.validate_on_submit(                    1899
    ):                                                                          1900
```

```
            name = request.form.get("name", "").strip()
            if not name:
                flash("指标名称不能为空", "error")
                return redirect(url_for("ahp.indicator_edit",
    code=code))
            ind.name = name
            ind.unit = request.form.get("unit", "").strip() or
    None
            ind.ideal_value = request.form.get("ideal_value",
    "").strip() or None
            ind.ideal_type = request.form.get("ideal_type") or
    ind.ideal_type
            units = request.form.getlist("enabled_units")
            ind.enabled_units = ",".join(units) if units else
    None
            ind.is_constraint = request.form.get("is_constraint"
    ) == "1"
            ind.description = request.form.get("description",
    "").strip() or None
            db.session.commit()
            flash("指标「%s」已保存" % ind.name, "ok")
            return redirect(url_for("ahp.indicators"))
    return render_template("ahp/indicator_edit.html",
    ind=ind,
                           unit_types=UNIT_TYPES,
                           type_label=ind_svc.ideal_type_lab
    el, form=form)
@bp.route("/indicators/reset", methods=["POST"])
def indicators_reset():
    """按论文取值重置指标定义。"""
    form = EmptyForm()
    if form.validate_on_submit():
        n = ind_svc.seed_indicators()
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")
    return redirect(url_for("ahp.indicators"))
# ---------------------------- 多专家矩阵 ----------------------------
    -----
def expert_scope():
    unit_type = request.values.get("unit_type") or
    UNIT_TYPES[0]
    level = int(request.values.get("level", 1))
    l1 = request.values.get("l1") or None
    return unit_type, level, l1
def expert_query(unit_type, level, l1):
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,
    level=level)
    if level == 2:
        q = q.filter_by(l1_name=l1)
    else:
        q = q.filter(ExpertMatrix.l1_name.is_(None))
```

```
    return q.order_by(ExpertMatrix.id)
def default_scheme(unit_type, level, l1=None):
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""
    name = "%s · %s%s" % (unit_type,
                          "一级指标" if level == 1 else "二级指标",
                          "（%s）" % l1 if l1 else "")
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          level=level,
    name=name).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    level=level, name=name,
                              is_default=False)
        db.session.add(scheme)
        db.session.flush()
    return scheme
@bp.route("/experts", methods=["GET", "POST"])
def experts():
    """多专家判断矩阵录入与汇总。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1"))
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
    == 2 else None),
```

```
                    expert_name=expert_name, level=level,
                    matrix_json=json.dumps([list(map(float,
    row)) for row in A]),
                    cr=round(r["cr"], 6)))
                db.session.commit()
                flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
    r["cr"]), "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
        db.session.commit()
        flash("已删除专家矩阵", "ok")
    return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                            level=level, l1=l1))
@bp.route("/experts/aggregate")
def experts_aggregate():
    """按几何平均汇总多专家矩阵，并做一致性检验。"""
    unit_type, level, l1 = expert_scope()
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    rows = expert_query(unit_type, level, l1).all()
    result = None
    if rows:
        mats = []
        for r in rows:
            try:
                mats.append([list(map(float, line))
                             for line in
    json.loads(r.matrix_json)])
            except Exception:
                continue
```

```
        if mats:
            agg = ahp.aggregate_matrices(mats)
            res = ahp.weights_and_consistency(agg)
            bad = ahp.find_most_inconsistent(agg)
            result = {
                "count": len(mats),
                "weights": [round(v, 4) for v in
    res["weights"]],
                "lambda_max": round(res["lambda_max"], 4),
                "ci": res["ci"], "cr": res["cr"],
                "consistent": res["consistent"],
                "most_inconsistent": bad,
                "expert_crs": [r.cr for r in rows if r.cr
    is not None],
            }
    return render_template("ahp/experts_aggregate.html",
    unit_type=unit_type,
                           level=level, l1=l1, names=names,
    rows=rows,
                           result=result)
# ==== app/views/indicator.py ====
# -*- coding: utf-8 -*-
"""M7 指标数据采集与管理 —— 路由与页面逻辑。
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标
（岩质边坡无土壤指标、平台无地形指标）。
"""
import io
import json
from flask import (Blueprint, render_template, request,
    redirect, url_for,
                   flash, send_file)
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_ENABLED_INDICATORS
from app.services import import_service as imp
bp = Blueprint("indicator", __name__,
    url_prefix="/indicator")
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)
SOIL_FIELDS = [
    ("soil_ratio", "土石比", ""),
    ("thickness", "土层厚度", "cm"),
    ("ph", "pH", ""),
    ("organic_matter", "有机质", "g/kg"),
]
VEG_FIELDS = [
    ("coverage", "植被覆盖度", "%"),
    ("shannon", "Shannon-Wiener 多样性指数", ""),
]
```

```
TERRAIN_FIELDS = [                                                  2101
    ("slope", "坡度", "°"),                                          2102
    ("aspect", "坡向", ""),                                          2103
]                                                                   2104
def unit_field_config(unit_type):                                   2105
    """根据单元类型返回需录入的字段分组。"""                                2106
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())            2107
    groups = []                                                     2108
    if "土壤指标" in enabled:                                         2109
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True    2110
    = 含重金属                                                       2111
    if "植被指标" in enabled:                                         2112
        groups.append(("植被指标", VEG_FIELDS, False))              2113
    if "地形指标" in enabled:                                         2114
        groups.append(("地形指标", TERRAIN_FIELDS, False))          2115
    return groups                                                   2116
@bp.route("/")                                                      2117
def index():                                                        2118
    patches = MinePatch.query.order_by(MinePatch.id.desc()).2119
    all()                                                           2120
    return render_template("indicator/index.html",                  2121
    patches=patches)                                                2122
@bp.route("/<int:unit_id>", methods=["GET", "POST"])                2123
def edit(unit_id):                                                  2124
    unit = db.get_or_404(MicroUnit, unit_id)                        2125
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir2126
    st()                                                            2127
    form = EmptyForm()                                              2128
    groups = unit_field_config(unit.unit_type)                      2129
    if request.method == "POST" and form.validate_on_submit(2130
    ):                                                              2131
        def f(name, cast=float):                                    2132
            v = request.form.get(name, "").strip()                  2133
            return cast(v) if v != "" else None                     2134
        heavy = {}                                                  2135
        for m in HEAVY_METALS:                                      2136
            v = f("hm_" + m)                                        2137
            if v is not None:                                       2138
                heavy[m] = v                                        2139
        if ind is None:                                             2140
            ind = IndicatorData(unit_id=unit.id)                    2141
            db.session.add(ind)                                     2142
        ind.soil_ratio = f("soil_ratio")                            2143
        ind.thickness = f("thickness")                              2144
        ind.ph = f("ph")                                            2145
        ind.organic_matter = f("organic_matter")                    2146
        ind.heavy_metals = json.dumps(heavy,                        2147
    ensure_ascii=False) if heavy else None                          2148
        ind.coverage = f("coverage")                                2149
        ind.shannon = f("shannon")                                  2150
```

```
        ind.slope = f("slope")
        ind.aspect = request.form.get("aspect", "").strip()
    or None
        db.session.commit()
        flash("指标数据已保存", "ok")
        return redirect(url_for("indicator.index"))
    # 组装当前值供模板回显
    current = {}
    if ind:
        current = {
            "soil_ratio": ind.soil_ratio, "thickness":
    ind.thickness,
            "ph": ind.ph, "organic_matter":
    ind.organic_matter,
            "coverage": ind.coverage, "shannon":
    ind.shannon,
            "slope": ind.slope, "aspect": ind.aspect,
        }
        current["heavy"] = ind.heavy_metals_dict()
    return render_template("indicator/edit.html",
                           unit=unit, form=form,
    groups=groups,
                           heavy_metals=HEAVY_METALS,
                           aspect_types=ASPECT_TYPES,
    current=current)
# ------------------------------ 批量导入 ------------------------------
    ----
@bp.route("/import", methods=["GET", "POST"])
def import_data():
    """批量导入指标数据（CSV / Excel）。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    form = EmptyForm()
    report = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        patch_id = request.form.get("patch_id", type=int)
        patch = db.session.get(MinePatch, patch_id) if
    patch_id else None
        if patch is None:
            flash("请选择矿山图斑", "error")
            return redirect(url_for("indicator.import_data")
    )
        file = request.files.get("datafile")
        if file is None or not file.filename:
            flash("请选择要导入的文件", "error")
            return redirect(url_for("indicator.import_data")
    )
        ext = file.filename.rsplit(".", 1)[-1].lower()
        if ext not in ("csv", "xlsx", "xls"):
```

```
            flash("仅支持 CSV 或 Excel(.xlsx/.xls)文件", "error")
            return redirect(url_for("indicator.import_data"))
)
        units = MicroUnit.query.filter_by(patch_id=patch.id)
.all()
        unit_map = {}
        for u in units:
            unit_map.setdefault(u.unit_type, u)
        if not unit_map:
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")
            return redirect(url_for("indicator.import_data"))
)
        try:
            parsed, errors = imp.parse_upload(file.filename,
file.stream, unit_map)
        except Exception as e:
            flash("文件解析失败：%s" % e, "error")
            return redirect(url_for("indicator.import_data"))
)
        saved = 0
        for rec in parsed:
            unit = unit_map[rec["unit_type"]]
            ind = IndicatorData.query.filter_by(unit_id=unit
.id).first()
            if ind is None:
                ind = IndicatorData(unit_id=unit.id)
                db.session.add(ind)
            f = rec["fields"]
            for key in ("soil_ratio", "thickness", "ph",
"organic_matter",
                        "coverage", "shannon", "slope",
"aspect"):
                if key in f:
                    setattr(ind, key, f[key])
            if rec["heavy_metals"]:
                ind.heavy_metals = json.dumps(rec["heavy_met
als"],

ensure_ascii=False)
            saved += 1
        db.session.commit()
        report = {"patch": patch.name, "saved": saved,
"errors": errors,
                  "total": len(parsed) + len(
                      [e for e in errors if e.startswith("第
")])}
        if saved:
            flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")
        else:
            flash("没有可导入的数据行，请检查文件内容", "error")
```

```
    return render_template("indicator/import.html",
    patches=patches,
                           form=form, report=report,
    headers=imp.HEADERS)
@bp.route("/import/template")
def import_template():
    """下载导入模板（CSV，含一行示例）。"""
    csv_text = imp.build_template_csv()
    data = csv_text.encode("utf-8-sig")  # 带 BOM，便于
    Excel 正确识别中文
    return send_file(io.BytesIO(data), as_attachment=True,
                     download_name="指标数据导入模板.csv",
                     mimetype="text/csv")
# ==== app/views/evaluate.py ====
# -*- coding: utf-8 -*-
"""M8 适宜性定量评价引擎 —— 路由与页面逻辑。
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。
"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData, EvaluationResult
from app.services import evaluation_service as ev
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all
    ()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind
    else None,
            "heavy_metals": ind.heavy_metals_dict() if ind
    else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else
```

```
    0.0,                                                                2301
            "data": data,                                               2302
        })                                                              2303
    return result                                                       2304
@bp.route("/")                                                          2305
def index():                                                            2306
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            2307
    all()                                                               2308
    # 每图斑是否已有评价结果                                            2309
    latest = {}                                                         2310
    for p in patches:                                                   2311
        r = EvaluationResult.query.filter_by(patch_id=p.id)             2312
    \                                                                   2313
                .order_by(EvaluationResult.id.desc()).first(            2314
    )                                                                   2315
        latest[p.id] = r                                                2316
    return render_template("evaluate/index.html",                       2317
    patches=patches, latest=latest)                                     2318
@bp.route("/<int:patch_id>", methods=["GET", "POST"])                   2319
def run(patch_id):                                                      2320
    patch = db.get_or_404(MinePatch, patch_id)                          2321
    form = EmptyForm()                                                  2322
    result = None                                                       2323
    if request.method == "POST" and form.validate_on_submit(            2324
    ):                                                                  2325
        units = load_patch_input(patch_id)                              2326
        if not units:                                                   2327
            flash("该图斑尚无微地质单元，请先划分微地质单元", "error")  2328
            return redirect(url_for("evaluate.index"))                  2329
        # 计算并入库                                                    2330
        res = ev.score_mine(units)                                      2331
        import json                                                     2332
        saved = EvaluationResult(                                       2333
            patch_id=patch.id,                                          2334
            final_score=round(res["final_score"], 4),                   2335
            grade=res["grade"],                                         2336
            measures=res["grade_info"]["measure"],                      2337
            unit_scores=json.dumps(                                     2338
                [{"unit_type": u["unit_type"], "ratio":                 2339
    u["ratio"],                                                         2340
                  "unit_score": round(u["unit_score"], 4)}              2341
                 for u in res["unit_results"]],                         2342
                ensure_ascii=False),                                    2343
        )                                                               2344
        db.session.add(saved)                                           2345
        db.session.commit()                                             2346
        return redirect(url_for("evaluate.result",                      2347
    result_id=saved.id))                                                2348
    return render_template("evaluate/run.html",                         2349
    patch=patch, form=form, result=result)                              2350
```

```
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    import json
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""M9 报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
from app.services import evaluation_service as ev
from app.services import report_service as rs
from app.utils import charts
from app.utils import browser
from app.views.evaluate import load_patch_input
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑、参照生态系统并执行评价，供明细页与导出复用。"""
    patch = db.get_or_404(MinePatch, patch_id)
    units = load_patch_input(patch_id)
    if not units:
        return patch, None, None, None
    result = ev.score_mine(units)
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, units
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
```

```
    return chart_units, chart_weight                                    2401
@bp.route("/")                                                          2402
def index():                                                            2403
    patches = MinePatch.query.order_by(MinePatch.id.desc()).            2404
    all()                                                               2405
    return render_template("report/index.html",                         2406
    patches=patches)                                                    2407
@bp.route("/<int:patch_id>")                                            2408
def detail(patch_id):                                                   2409
    patch, reference, result, units = _prepare(patch_id)                2410
    if result is None:                                                  2411
        flash("该图斑尚无微地质单元，无法生成报告", "error")                2412
        return redirect(url_for("report.index"))                        2413
    chart_units, chart_weight = _charts(result)                         2414
    return render_template("report/detail.html",                        2415
    patch=patch, result=result,                                         2416
                                 reference=reference,                   2417
    chart_units=chart_units,                                            2418
                                 chart_weight=chart_weight)             2419
@bp.route("/<int:patch_id>/word")                                       2420
def word(patch_id):                                                     2421
    patch, reference, result, units = _prepare(patch_id)                2422
    if result is None:                                                  2423
        flash("该图斑尚无微地质单元，无法生成报告", "error")                2424
        return redirect(url_for("report.index"))                        2425
    buf = rs.build_report(patch, result)                                2426
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name              2427
    return send_file(buf, as_attachment=True,                           2428
    download_name=filename,                                             2429
                         mimetype="application/vnd.openxmlformat        2430
    s-officedocument."                                                  2431
                                   "wordprocessingml.document")         2432
@bp.route("/<int:patch_id>/pdf")                                        2433
def pdf(patch_id):                                                      2434
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""                2435
    patch, reference, result, units = _prepare(patch_id)                2436
    if result is None:                                                  2437
        flash("该图斑尚无微地质单元，无法生成报告", "error")                2438
        return redirect(url_for("report.index"))                        2439
    chart_units, chart_weight = _charts(result)                         2440
    html = render_template(                                             2441
        "report/print.html", patch=patch,                               2442
    reference=reference, result=result,                                 2443
        chart_units=chart_units, chart_weight=chart_weight,             2444
        generated_at=datetime.now().strftime("%Y-%m-%d                  2445
    %H:%M"))                                                            2446
    data, err = browser.html_to_pdf(html)                               2447
    if err:                                                             2448
        flash("PDF 导出失败：%s" % err, "error")                         2449
        return redirect(url_for("report.detail",                        2450
```

```
        patch_id=patch_id))
        return send_file(io.BytesIO(data), as_attachment=True,
                         download_name="自然恢复适宜性评价报告_%s.pdf" %
        patch.name,
                         mimetype="application/pdf")
# ==== app/templates/base.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
    initial-scale=1.0">
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock
    %}</title>
  <style>
    /* ---------- 设计变量：单一强调色 + 中性灰阶 ---------- */
    :root {
      --accent: #1f3a5f;
      --accent-hover: #2a4d7c;
      --accent-soft: #eef2f7;
      --danger: #b3261e;
      --danger-soft: #fdecea;
      --ok: #1e6b3a;
      --ok-soft: #eaf6ef;
      --ink: #23262b;
      --ink-soft: #6b7280;
      --line: #dfe3e8;
      --surface: #ffffff;
      --canvas: #f5f6f8;
      --radius: 6px;
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);
      --header-h: 56px;
      --nav-w: 208px;
    }
    * { box-sizing: border-box; }
    html, body { margin: 0; padding: 0; }
    body {
      font-family: "Microsoft YaHei", "PingFang SC",
    "SimSun", sans-serif;
      font-size: 14px;
      line-height: 1.6;
      color: var(--ink);
      background: var(--canvas);
    }
    /* ---------- 跳到主要内容（键盘可达性） ---------- */
    .skip-link {
      position: absolute;
      left: 12px;
      top: -60px;
      z-index: 30;
```

```
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 顶部标题栏 ---------- */
.app-header {
    display: flex;
    align-items: center;
    gap: 12px;
    height: var(--header-h);
    padding: 0 20px;
    background: var(--accent);
    color: #fff;
}
.app-header__title {
    font-size: 16px;
    font-weight: 600;
    letter-spacing: .2px;
}
.app-header__version {
    padding: 1px 8px;
    font-size: 12px;
    border: 1px solid rgba(255, 255, 255, .45);
    border-radius: 10px;
    font-variant-numeric: tabular-nums;
}
.app-header__spacer { flex: 1; }
.app-header__user {
    display: flex;
    align-items: center;
    gap: 10px;
    font-size: 13px;
}
.app-header__user a {
    color: #d6e3f4;
    text-decoration: none;
    padding: 4px 8px;
    border-radius: 4px;
}
.app-header__user a:hover,
.app-header__user a:focus-visible {
    color: #fff;
    background: rgba(255, 255, 255, .14);
}
/* ---------- 主体：左侧导航 + 内容区 ---------- */
.app-body {
```

```
    display: flex;
    align-items: stretch;
    min-height: calc(100dvh - var(--header-h));
}
.app-nav {
    flex: 0 0 var(--nav-w);
    width: var(--nav-w);
    padding: 14px 10px;
    background: var(--surface);
    border-right: 1px solid var(--line);
}
.app-nav__group {
    margin: 0 0 4px;
    padding: 10px 10px 6px;
    font-size: 12px;
    color: var(--ink-soft);
}
.app-nav ul {
    margin: 0;
    padding: 0;
    list-style: none;
}
.app-nav a {
    display: block;
    padding: 9px 12px;
    margin-bottom: 2px;
    border-radius: var(--radius);
    color: var(--ink);
    text-decoration: none;
    border-left: 3px solid transparent;
}
.app-nav a:hover { background: var(--accent-soft); }
.app-nav a:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
}
.app-nav a[aria-current="page"] {
    background: var(--accent-soft);
    border-left-color: var(--accent);
    color: var(--accent);
    font-weight: 600;
}
.app-main {
    flex: 1;
    min-width: 0;
    padding: 22px 26px 40px;
}
.container { max-width: 1040px; }
/* ---------- 卡片 ---------- */
.card {
```

```
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
```

```
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
    width: 104px;
    text-align: center;
    font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
    border-color: var(--accent);
}
.checkbox {
    display: inline-flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;
    user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ---------- */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 8px 18px;
    font-family: inherit;
    font-size: 14px;
    line-height: 1.4;
    color: #fff;
    background: var(--accent);
```

```
    border: 1px solid transparent;
    border-radius: 4px;
    cursor: pointer;
    text-decoration: none;
}
.btn:hover { background: var(--accent-hover); }
.btn:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.btn--ghost {
    color: var(--ink);
    background: #fff;
    border-color: #c6ccd4;
}
.btn--ghost:hover { background: #f1f3f6; }
.btn--danger {
    color: var(--danger);
    background: #fff;
    border-color: #e6b4b0;
}
.btn--danger:hover { background: var(--danger-soft); }
.btn--sm { padding: 4px 12px; font-size: 13px; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
.actions { display: flex; flex-wrap: wrap; align-items:
center; gap: 8px; }
.actions form { display: inline; }
/* ---------- 提示条 ---------- */
.flash {
    padding: 10px 14px;
    margin-bottom: 14px;
    border-radius: var(--radius);
    border-left: 4px solid;
}
.flash.error {
    color: var(--danger);
    background: var(--danger-soft);
    border-left-color: var(--danger);
}
.flash.ok {
    color: var(--ok);
    background: var(--ok-soft);
    border-left-color: var(--ok);
}
/* ---------- 文字工具类 ---------- */
.muted { color: var(--ink-soft); font-size: 12.5px; }
.mono { font-family: Consolas, "Courier New",
monospace; font-variant-numeric: tabular-nums; }
.num { font-variant-numeric: tabular-nums; }
.warn { color: var(--danger); font-weight: 600; }
.ok { color: var(--ok); font-weight: 600; }
.nowrap { white-space: nowrap; }
```

```
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
.login-wrap {
  display: flex;
  justify-content: center;
  padding: 64px 20px;
}
.login-card { width: 100%; max-width: 380px; }
.login-card .field__label { font-size: 14px; color:
var(--ink); }
/* ---------- 首页功能块 ---------- */
.tile-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(232px, 1fr));
  gap: 12px;
}
.tile {
  display: block;
  padding: 16px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  color: var(--ink);
  text-decoration: none;
}
.tile:hover { border-color: var(--accent); background:
#fbfcfe; }
.tile:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.tile__title { font-weight: 600; color: var(--accent);
margin-bottom: 5px; }
.tile__desc { font-size: 12.5px; color:
var(--ink-soft); text-wrap: pretty; }
```

```
      /* ---------- 结果摘要 ---------- */
      .stat-row {
        display: flex;
        flex-wrap: wrap;
        gap: 28px;
        margin: 4px 0 6px;
      }
      .stat__label { font-size: 12.5px; color: var(--ink-soft); }
      .stat__value { font-size: 20px; font-weight: 700; font-variant-numeric: tabular-nums; }
      /* ---------- 窄屏适配 ---------- */
      @media (max-width: 860px) {
        .app-body { flex-direction: column; }
        .app-nav {
          flex: none;
          width: auto;
          border-right: none;
          border-bottom: 1px solid var(--line);
          padding: 8px 10px;
        }
        .app-nav ul { display: flex; flex-wrap: wrap; gap: 4px; }
        .app-nav a {
          margin-bottom: 0;
          border-left: none;
          border-bottom: 3px solid transparent;
        }
        .app-nav a[aria-current="page"] { border-left: none; border-bottom-color: var(--accent); }
        .app-nav__group { display: none; }
        .app-main { padding: 16px 14px 32px; }
      }
    </style>
</head>
<body>
    <a class="skip-link" href="#main">跳到主要内容</a>
    <header class="app-header">
      <span class="app-header__title">历史遗留废弃矿山自然恢复适宜性评价系统</span>
      <span class="app-header__version">V1.0</span>
      <span class="app-header__spacer"></span>
      {% if current_user.is_authenticated %}
      <span class="app-header__user">
        <span>{{ current_user.username }} ({{ current_user.role }})</span>
        <a href="/logout">退出</a>
      </span>
      {% endif %}
    </header>
```

```
  <div class="app-body">
    {% if current_user.is_authenticated %}
    <nav class="app-nav" aria-label="功能导航">
      <p class="app-nav__group">功能模块</p>
      <ul>
        {% for label, href, active in nav_items %}
        <li>
          <a href="{{ href }}"{% if active %}
    aria-current="page"{% endif %}>{{ label }}</a>
        </li>
        {% endfor %}
      </ul>
    </nav>
    {% endif %}
    <main id="main" class="app-main">
      <div class="container">
        {% with messages = get_flashed_messages(with_categor
    ies=true) %}
          {% for category, message in messages %}
          <p class="flash {{ category }}" role="status">{{
    message }}</p>
          {% endfor %}
        {% endwith %}
        {% block content %}{% endblock %}
      </div>
    </main>
  </div>
</body>
</html>
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="login-wrap">
  <div class="card login-card">
    <h2>用户登录</h2>
    <form method="post" novalidate>
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="{{
    form.username.id }}">用户名</label>
        {{ form.username(id=form.username.id,
    autocomplete="username") }}
      </div>
      <div class="field">
        <label class="field__label" for="{{
    form.password.id }}">密码</label>
        {{ form.password(id=form.password.id,
    autocomplete="current-password") }}
      </div>
```

```
        <div class="field">
          <label class="checkbox" for="{{ form.remember.id
    }}">
            {{ form.remember(id=form.remember.id) }}
            <span>记住我（下次自动登录，一个月内有效）</span>
          </label>
        </div>
        <p style="margin:18px 0 0;">
          {{ form.submit(class_="btn", style="width:100%;") }}
        </p>
      </form>
      <p class="muted" style="margin:14px 0 0;">演示账号：admin /
      admin123</p>
    </div>
</div>
{% endblock %}
# ==== app/templates/auth/home.html ====
{% extends "base.html" %}
{% block title %}首页{% endblock %}
{% block content %}
<div class="card">
  <h2>系统功能</h2>
  <p class="muted">
    本系统以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》
    建立的评价方法为算法内核，覆盖从基础档案到评价报告输出的完整流程。
  </p>
  <div class="tile-grid" style="margin-top:16px;">
    <a class="tile" href="/project/">
      <div class="tile__title">M2 项目与矿山档案</div>
      <div class="tile__desc">建立评价项目、录入矿山图斑、设置参照生态系统</div>
    </a>
    <a class="tile" href="/constraint/">
      <div class="tile__title">M4 限制性指标判定</div>
      <div class="tile__desc">政策、区位约束、土壤污染、地质灾害、耕地属性一票否决</div>
    </a>
    <a class="tile" href="/standard/">
      <div class="tile__title">M3 政策与标准知识库</div>
      <div class="tile__desc">11 项地方标准与政策文件、省际口径比对</div>
    </a>
    <a class="tile" href="/ahp/">
      <div class="tile__title">M6 指标体系与 AHP 权重</div>
      <div class="tile__desc">判断矩阵录入、权重计算、一致性检验</div>
    </a>
    <a class="tile" href="/indicator/">
      <div class="tile__title">M7 指标数据采集</div>
      <div class="tile__desc">按微地质单元录入土壤、植被、地形指标</div>
    </a>
    <a class="tile" href="/evaluate/">
      <div class="tile__title">M8 适宜性定量评价</div>
```

```
        <div class="tile__desc">标准化、分单元赋分、面积加权、等级判定</div>
      </a>
      <a class="tile" href="/report/">
        <div class="tile__title">M9 报告与结果输出</div>
        <div class="tile__desc">得分明细、Word 评价报告导出</div>
      </a>
      <a class="tile" href="/users">
        <div class="tile__title">M1 用户与权限管理</div>
        <div class="tile__desc">用户维护与角色权限</div>
      </a>
    </div>
</div>
{% endblock %}
# ==== app/templates/auth/users.html ====
{% extends "base.html" %}
{% block title %}用户与权限管理{% endblock %}
{% block content %}
<div class="card">
    <h2>用户管理</h2>
    <p class="muted">
      系统设有管理员、评价员、查看者三种角色：管理员可维护用户与基础数据，
      评价员可建项目、录数据、执行评价，查看者仅可查阅。
    </p>
    <p class="actions" style="margin-bottom:12px;">
      {% if current_user.role == "admin" %}
      <a class="btn btn--sm" href="/users/add">新增用户</a>
      {% endif %}
      <a class="btn btn--sm btn--ghost"
      href="/logs">查看操作日志</a>
    </p>
    {% if current_user.role != "admin" %}
    <p class="muted">
      当前角色为「{{ role_labels.get(current_user.role,
      current_user.role) }}」，
      仅管理员可新增、编辑或删除用户。
    </p>
    {% endif %}
    <table>
      <caption class="muted">共 {{ users|length }}
      个用户</caption>
      <thead>
        <tr>
          <th scope="col">ID</th>
          <th scope="col">用户名</th>
          <th scope="col">角色</th>
          <th scope="col">状态</th>
          {% if current_user.role == "admin" %}<th
      scope="col">操作</th>{% endif %}
        </tr>
      </thead>
```

```
      <tbody>                                                                  3001
        {% for u in users %}                                                   3002
        <tr>                                                                   3003
          <td class="num">{{ u.id }}</td>                                      3004
          <td>                                                                 3005
            {{ u.username }}                                                   3006
            {% if u.id == current_user.id %}<span                              3007
      class="muted">（当前登录）</span>{% endif %}                                3008
          </td>                                                                3009
          <td>{{ role_labels.get(u.role, u.role) }}</td>                       3010
          <td>                                                                 3011
            {% if u.active %}<span class="tag                                  3012
      tag--ok">启用</span>                                                     3013
            {% else %}<span class="tag tag--muted">停用</span>{%               3014
      endif %}                                                                 3015
          </td>                                                                3016
          {% if current_user.role == "admin" %}                                3017
          <td class="nowrap">                                                  3018
            <a href="/users/{{ u.id }}/edit">编辑</a>                          3019
             |                                                       3020
            <form method="post" action="/users/{{ u.id                         3021
      }}/delete"                                                               3022
                  onsubmit="return confirm('确定删除用户「{{                   3023
      u.username }}」吗？\n该操作无法恢复。');">                                  3024
              {{ form.hidden_tag() }}                                          3025
              <button type="submit" class="btn btn--sm                         3026
      btn--danger"                                                             3027
                      {% if u.id == current_user.id %}disabled                 3028
                      title="不能删除当前登录的账号"{% endif                     3029
      %}>删除</button>                                                         3030
            </form>                                                            3031
          </td>                                                                3032
          {% endif %}                                                          3033
        </tr>                                                                  3034
        {% endfor %}                                                           3035
      </tbody>                                                                 3036
    </table>                                                                   3037
</div>                                                                         3038
{% endblock %}                                                                 3039
# ==== app/templates/auth/add_user.html ====                                   3040
{% extends "base.html" %}                                                      3041
{% block title %}新增用户{% endblock %}                                        3042
{% block content %}                                                            3043
<div class="card" style="max-width:560px;">                                    3044
  <h2>新增用户</h2>                                                            3045
  <form method="post">                                                         3046
    {{ form.hidden_tag() }}                                                    3047
    <div class="field">                                                        3048
      <label class="field__label" for="username">用户名</label>                3049
      <input id="username" type="text" name="username"                         3050
```

```
      required                                                          3051
                autocomplete="off" style="width:280px;">                3052
      </div>                                                            3053
      <div class="field">                                               3054
        <label class="field__label" for="password">密码</label>         3055
        <input id="password" type="password" name="password"            3056
      required                                                          3057
                autocomplete="new-password"                             3058
      style="width:280px;">                                             3059
      </div>                                                            3060
      <div class="field">                                               3061
        <label class="field__label" for="role">角色</label>             3062
        <select id="role" name="role">                                  3063
          <option value="viewer">查看者（仅可查阅）</option>              3064
          <option value="evaluator">评价员（可建项目、录数据、执行评价）</option3065
      >                                                                 3066
          <option value="admin">管理员（可维护用户与基础数据）</option>   3067
        </select>                                                       3068
      </div>                                                            3069
      <p class="actions" style="margin-top:18px;">                      3070
        <button type="submit" class="btn">保存</button>                 3071
        <a class="btn btn--ghost" href="/users">返回</a>                3072
      </p>                                                              3073
    </form>                                                             3074
</div>                                                                  3075
{% endblock %}                                                          3076
# ==== app/templates/project/index.html ====                            3077
{% extends "base.html" %}                                               3078
{% block title %}项目与矿山档案{% endblock %}                           3079
{% block content %}                                                     3080
<div class="card">                                                      3081
  <h2>评价项目</h2>                                                     3082
  <p class="actions" style="margin-bottom:12px;">                       3083
    <a class="btn btn--sm" href="/project/add">新建项目</a>             3084
  </p>                                                                  3085
  {% if not projects %}                                                 3086
  <p class="muted">                                                     3087
    暂无项目。点击上方「新建项目」开始建立评价项目，                   3088
    或运行 <span class="mono">python seed_demo.py</span>               3089
    载入论文案例。                                                      3090
  </p>                                                                  3091
  {% else %}                                                            3092
  <table>                                                               3093
    <caption class="muted">共 {{ projects|length }}                     3094
    个项目</caption>                                                    3095
    <thead>                                                             3096
      <tr>                                                              3097
        <th scope="col">项目名称</th>                                   3098
        <th scope="col">编号</th>                                       3099
        <th scope="col">状态</th>                                       3100
```

```
            <th scope="col">图斑数</th>                                    3101
            <th scope="col">操作</th>                                      3102
          </tr>                                                            3103
        </thead>                                                           3104
        <tbody>                                                            3105
          {% for p in projects %}                                          3106
          <tr>                                                             3107
            <td style="text-align:left;">{{ p.name }}</td>                 3108
            <td class="mono">{{ p.code or "—" }}</td>                      3109
            <td><span class="tag tag--muted">{{ p.status                   3110
        }}</span></td>                                                     3111
            <td class="num">{{ p.patches.count() }}</td>                   3112
            <td class="nowrap">                                            3113
              <a href="/project/{{ p.id }}/patches">矿山图斑</a>           3114
               | <a href="/project/{{ p.id                       3115
        }}/edit">编辑</a>                                                  3116
               |                                                 3117
              <form method="post" action="/project/{{ p.id                 3118
        }}/delete"                                                         3119
                    onsubmit="return confirm('确定删除项目「{{ p.name      3120
        }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  3121
                {{ form.hidden_tag() }}                                    3122
                <button type="submit" class="btn btn--sm                   3123
        btn--danger">删除</button>                                         3124
              </form>                                                      3125
            </td>                                                          3126
          </tr>                                                            3127
          {% endfor %}                                                     3128
        </tbody>                                                           3129
      </table>                                                             3130
      {% endif %}                                                          3131
    </div>                                                                 3132
    {% endblock %}                                                         3133
    # ==== app/templates/project/edit.html ====                            3134
    {% extends "base.html" %}                                              3135
    {% block title %}{{ "编辑项目" if project else "新建项目" }}{%          3136
        endblock %}                                                        3137
    {% block content %}                                                    3138
    <div class="card" style="max-width:640px;">                            3139
      <h2>{{ "编辑项目" if project else "新建项目" }}</h2>                 3140
      <form method="post">                                                 3141
        {{ form.hidden_tag() }}                                            3142
        <div class="field">                                                3143
          <label class="field__label" for="name">项目名称</label>          3144
          <input id="name" type="text" name="name"                         3145
        style="width:100%;" required                                       3146
                 value="{{ project.name if project else '' }}">            3147
        </div>                                                             3148
        <div class="field">                                                3149
          <label class="field__label" for="code">项目编号</label>          3150
```

```
        <input id="code" type="text" name="code"                    3151
      style="width:240px;"                                          3152
             value="{{ project.code or '' if project else           3153
      '' }}">                                                       3154
      </div>                                                        3155
      {% if project %}                                              3156
      <div class="field">                                           3157
        <label class="field__label" for="status">状态</label>       3158
        <select id="status" name="status">                          3159
          {% for s in ["进行中", "已完成", "已归档"] %}             3160
          <option value="{{ s }}" {% if project.status == s         3161
      %}selected{% endif %}>{{ s }}</option>                        3162
          {% endfor %}                                              3163
        </select>                                                   3164
      </div>                                                        3165
      {% endif %}                                                   3166
      <div class="field">                                           3167
        <label class="field__label" for="description">项目说明</la3168
      bel>                                                          3169
        <textarea id="description" name="description">{{          3170
      project.description or '' if project else ''                  3171
      }}</textarea>                                                 3172
      </div>                                                        3173
      <p class="actions" style="margin-top:18px;">                  3174
        <button type="submit" class="btn">保存</button>             3175
        <a class="btn btn--ghost" href="/project/">返回</a>         3176
      </p>                                                          3177
    </form>                                                         3178
</div>                                                              3179
{% endblock %}                                                      3180
# ==== app/templates/project/patches.html ====                      3181
{% extends "base.html" %}                                           3182
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}        3183
{% block content %}                                                 3184
<div class="card">                                                  3185
  <h2>矿山图斑 · {{ project.name }}</h2>                            3186
  <p class="actions" style="margin-bottom:12px;">                   3187
    <a class="btn btn--sm" href="/project/{{ project.id             3188
    }}/patch/add">新建图斑</a>                                      3189
    <a class="btn btn--sm btn--ghost"                               3190
    href="/project/">返回项目列表</a>                               3191
  </p>                                                              3192
  {% if not project.patches.count() %}                              3193
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>  3194
  {% else %}                                                        3195
  <table>                                                           3196
    <caption class="muted">共 {{ project.patches.count() }}         3197
    个图斑</caption>                                                3198
    <thead>                                                         3199
      <tr>                                                          3200
```

```
            <th scope="col">图斑名称</th>                                    3201
            <th scope="col">行政区</th>                                      3202
            <th scope="col">面积（公顷）</th>                                3203
            <th scope="col">开采矿种</th>                                    3204
            <th scope="col">微地质单元</th>                                  3205
            <th scope="col">操作</th>                                        3206
          </tr>                                                              3207
        </thead>                                                             3208
        <tbody>                                                              3209
          {% for p in project.patches %}                                     3210
          <tr>                                                               3211
            <td style="text-align:left;">{{ p.name }}</td>                   3212
            <td>{{ p.admin_region or "—" }}</td>                             3213
            <td class="num">{{ p.area_ha if p.area_ha is not                 3214
        none else "—" }}</td>                                                3215
            <td>{{ p.mineral_type or "—" }}</td>                             3216
            <td class="num">{{ p.units.count() }}</td>                       3217
            <td class="nowrap">                                              3218
              <a href="/unit/{{ p.id }}">微地质单元</a>                      3219
               | <a href="/constraint/{{ p.id                      3220
        }}">限制性判定</a>                                                   3221
               | <a href="/project/patch/{{ p.id                   3222
        }}/edit">编辑</a>                                                    3223
               |                                                   3224
              <form method="post" action="/project/patch/{{                  3225
        p.id }}/delete"                                                      3226
                    onsubmit="return confirm('确定删除图斑「{{ p.name        3227
        }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  3228
                {{ form.hidden_tag() }}                                      3229
                <button type="submit" class="btn btn--sm                     3230
        btn--danger">删除</button>                                           3231
              </form>                                                        3232
            </td>                                                            3233
          </tr>                                                              3234
          {% endfor %}                                                       3235
        </tbody>                                                             3236
      </table>                                                               3237
      {% endif %}                                                            3238
    </div>                                                                   3239
    {% endblock %}                                                           3240
    # ==== app/templates/project/patch_edit.html ====                        3241
    {% extends "base.html" %}                                                3242
    {% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%             3243
        endblock %}                                                          3244
    {% block content %}                                                      3245
    <div class="card" style="max-width:720px;">                              3246
      <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name        3247
        }}</h2>                                                              3248
      <form method="post">                                                   3249
        {{ form.hidden_tag() }}                                              3250
```

```
<div class="field">
  <label class="field__label" for="patch-name">图斑名称</label>
  <input id="patch-name" type="text" name="name" style="width:100%;" required
         value="{{ patch.name if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="admin-region">所在行政区</label>
  <input id="admin-region" type="text" name="admin_region" style="width:100%;"
         value="{{ patch.admin_region or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="area-ha">图斑面积（公顷）</label>
  <input id="area-ha" type="number" step="any" name="area_ha"
         value="{{ patch.area_ha if patch and patch.area_ha is not none else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</label>
  <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
         value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox"
```

```
      name="responsible_entity_missing" value="1"                    3301
                  {% if not patch or patch.responsible_entity_m      3302
      issing %}checked{% endif %}>                                   3303
          <span>责任主体缺失</span>                                   3304
        </label>                                                     3305
      </div>                                                         3306
      <h3>参照生态系统（指标标准化基准）</h3>                          3307
      <p class="muted">                                              3308
        论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。  3309
        该基准与图斑绑定管理，可按不同参照生态系统分别设定。          3310
      </p>                                                           3311
      <div class="field">                                            3312
        <label class="field__label" for="coverage-benchmark">植      3313
      被覆盖度基准（%）</label>                                      3314
        <input id="coverage-benchmark" type="number"                 3315
      step="any" name="coverage_benchmark"                           3316
                value="{{ reference.coverage_benchmark if            3317
      reference else 75 }}">                                         3318
      </div>                                                         3319
      <div class="field">                                            3320
        <label class="field__label" for="shannon-benchmark">Sh       3321
      annon-Wiener 多样性指数基准</label>                            3322
        <input id="shannon-benchmark" type="number"                  3323
      step="any" name="shannon_benchmark"                            3324
                value="{{ reference.shannon_benchmark if             3325
      reference else 0.7 }}">                                        3326
      </div>                                                         3327
      <div class="field">                                            3328
        <label class="field__label" for="dominant-species">建群种    3329
      </label>                                                       3330
        <input id="dominant-species" type="text"                     3331
      name="dominant_species" style="width:100%;"                    3332
                value="{{ reference.dominant_species or '' if        3333
      reference else '' }}">                                         3334
      </div>                                                         3335
      <p class="actions" style="margin-top:18px;">                   3336
        <button type="submit" class="btn">保存</button>              3337
        <a class="btn btn--ghost" href="/project/{{                  3338
      project.id }}/patches">返回</a>                                3339
      </p>                                                           3340
    </form>                                                          3341
</div>                                                               3342
{% if patch %}                                                       3343
<div class="card" style="max-width:720px;">                          3344
    <h2>遥感影像挂接</h2>                                            3345
    <p class="muted">                                                3346
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。  3347
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。    3348
    </p>                                                             3349
    {% if patch.image_path %}                                        3350
```

```
  <figure style="margin:0 0 16px;">                                    3351
    <img src="{{ url_for('static', filename=patch.image_path3352
    ) }}"                                                              3353
         alt="图斑「{{ patch.name }}」已挂接的遥感影像"                  3354
         style="max-width:100%;border:1px solid                        3355
    var(--line);border-radius:6px;">                                   3356
    <figcaption class="muted" style="margin-top:6px;">                 3357
      {{ patch.image_note or "遥感影像" }}                             3358
      {% if patch.image_uploaded_at %}                                 3359
      · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d           3360
    %H:%M") }}                                                         3361
      {% endif %}                                                      3362
    </figcaption>                                                      3363
  </figure>                                                            3364
  <form method="post" action="/project/patch/{{ patch.id               3365
    }}/image/delete"                                                   3366
        onsubmit="return confirm('确定移除该遥感影像吗？');">          3367
    {{ form.hidden_tag() }}                                            3368
    <button type="submit" class="btn btn--sm                           3369
    btn--danger">移除影像</button>                                     3370
  </form>                                                              3371
  <hr style="margin:16px 0;border:none;border-top:1px solid            3372
    var(--line);">                                                     3373
  {% endif %}                                                          3374
  <form method="post" action="/project/patch/{{ patch.id               3375
    }}/image"                                                          3376
        enctype="multipart/form-data">                                 3377
    {{ form.hidden_tag() }}                                            3378
    <div class="field">                                                3379
      <label class="field__label" for="image-file">                    3380
        {{ "替换影像文件" if patch.image_path else "选择影像文件" }}   3381
      </label>                                                         3382
      <input id="image-file" type="file" name="image"                  3383
             accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t3384
    iff" required>                                                     3385
    </div>                                                             3386
    <div class="field">                                                3387
      <label class="field__label" for="image-note">影像说明（如来源、3388
    时相）</label>                                                     3389
      <input id="image-note" type="text" name="image_note"             3390
    style="width:100%;"                                                3391
             placeholder="如：2025 年 6 月无人机正射影像"              3392
             value="{{ patch.image_note or '' }}">                     3393
    </div>                                                             3394
    <p class="actions">                                                3395
      <button type="submit" class="btn">{{ "替换影像" if               3396
    patch.image_path else "上传并挂接" }}</button>                     3397
    </p>                                                               3398
  </form>                                                              3399
</div>                                                                 3400
```

```
{% endif %}
{% endblock %}
# ==== app/templates/unit/index.html ====
{% extends "base.html" %}
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元 · {{ patch.name }}</h2>
  <p class="muted">
    按论文 4.1 节将矿山图斑划分为岩质边坡、碎石边坡、平台三类微地质单元。
    面积占比由面积自动计算，即论文式（9） 中的权重系数。
  </p>
  {% if units %}
  <table>
    <caption class="muted">
      共 {{ units|length }} 个单元；
      面积合计 <span class="num">{{ "%.4f"|format(total_area)
    }}</span>；
      占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *
    100) }}</span>
      {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >
    1.001) %}
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <a href="/unit/edit/{{ u.id }}">编辑</a>
           | 
          <form method="post" action="/unit/delete/{{ u.id
```

```
      }}"                                                                    3451
                      onsubmit="return confirm('确定删除微地质单元「{{      3452
      u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">  3453
              {{ form.hidden_tag() }}                                        3454
              <button type="submit" class="btn btn--sm                       3455
      btn--danger">删除</button>                                             3456
            </form>                                                          3457
          </td>                                                              3458
        </tr>                                                                3459
        {% endfor %}                                                         3460
      </tbody>                                                               3461
    </table>                                                                 3462
    {% else %}                                                               3463
    <p class="muted">尚未划分微地质单元。请在下方添加。</p>                  3464
    {% endif %}                                                              3465
    <h3>添加微地质单元</h3>                                                  3466
    <form method="post" action="/unit/{{ patch.id }}/add">                   3467
      {{ form.hidden_tag() }}                                                3468
      <div class="actions">                                                  3469
        <label class="field__label" for="add-unit-type"                      3470
      style="margin:0;">类型</label>                                         3471
        <select id="add-unit-type" name="unit_type">                         3472
          {% for t in unit_types %}<option value="{{ t }}">{{                3473
      t }}</option>{% endfor %}                                              3474
        </select>                                                            3475
        <label class="field__label" for="add-unit-area"                      3476
      style="margin:0;">面积</label>                                         3477
        <input id="add-unit-area" type="number" step="any"                   3478
      name="area">                                                           3479
        <label class="field__label" for="add-unit-desc"                      3480
      style="margin:0;">说明</label>                                         3481
        <input id="add-unit-desc" type="text"                                3482
      name="description" style="width:220px;">                               3483
        <button type="submit" class="btn btn--sm">添加</button>              3484
      </div>                                                                 3485
    </form>                                                                  3486
    <p class="actions" style="margin-top:20px;">                             3487
      <a class="btn btn--sm btn--ghost" href="/project/{{                    3488
      patch.project_id }}/patches">返回图斑列表</a>                          3489
      <a class="btn btn--sm btn--ghost"                                      3490
      href="/indicator/">去录入指标数据</a>                                  3491
    </p>                                                                     3492
</div>                                                                       3493
{% endblock %}                                                               3494
# ==== app/templates/unit/edit.html ====                                     3495
{% extends "base.html" %}                                                    3496
{% block title %}编辑微地质单元{% endblock %}                                3497
{% block content %}                                                          3498
<div class="card">                                                           3499
  <h2>编辑微地质单元 · {{ patch.name }}</h2>                                 3500
```

```
  <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type">
        {% for t in unit_types %}
        <option value="{{ t }}" {% if unit.unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="unit-area">面积</label>
      <input id="unit-area" type="number" step="any" name="area"
             value="{{ unit.area if unit.area is not none else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit-desc">说明</label>
      <input id="unit-desc" type="text" name="description" style="width:420px;"
             value="{{ unit.description or '' }}">
    </div>
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/unit/{{ unit.patch_id }}">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>
  <p class="muted">
    收录论文表　1　梳理的地方标准与政策文件，并将其判定条件结构化为可执行规则，
    支持同一指标在不同省份的阈值横向比对。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
```

```
{% if local %}
<table>
  <thead>
    <tr>
      <th scope="col">省份</th>
      <th scope="col">名称</th>
      <th scope="col">编号</th>
      <th scope="col">规则数</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for s in local %}
    <tr>
      <td>{{ s.province }}</td>
      <td style="text-align:left;">{{ s.name }}</td>
      <td class="mono">{{ s.doc_number or "—" }}</td>
      <td class="num">{{ s.rules|length }}</td>
      <td class="nowrap">
        <a href="/standard/{{ s.id }}/rules">规则</a>
         | <a href="/standard/{{ s.id }}/edit">编辑</a>
         | 
        <form method="post" action="/standard/{{ s.id }}/delete"
              onsubmit="return confirm('确定删除「{{ s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length }} 条规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
{% else %}
<p class="muted">暂无地方标准。运行 <span class="mono">python seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>
{% endif %}
<h3>政策文件</h3>
{% if policy %}
<table>
  <thead>
    <tr>
      <th scope="col">省份</th>
      <th scope="col">名称</th>
      <th scope="col">文号</th>
      <th scope="col">规则数</th>
```

```
            <th scope="col">操作</th>                                                    3601
          </tr>                                                                          3602
        </thead>                                                                         3603
        <tbody>                                                                          3604
          {% for s in policy %}                                                          3605
          <tr>                                                                           3606
            <td>{{ s.province }}</td>                                                    3607
            <td style="text-align:left;">{{ s.name }}</td>                               3608
            <td class="mono">{{ s.doc_number or "—" }}</td>                              3609
            <td class="num">{{ s.rules|length }}</td>                                    3610
            <td class="nowrap">                                                          3611
              <a href="/standard/{{ s.id }}/rules">规则</a>                              3612
               | <a href="/standard/{{ s.id                                    3613
        }}/edit">编辑</a>                                                                3614
               |                                                               3615
              <form method="post" action="/standard/{{ s.id                              3616
        }}/delete"                                                                       3617
                    onsubmit="return confirm('确定删除「{{                               3618
        s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length                   3619
        }} 条规则将一并删除，且无法恢复。');">                                           3620
                {{ form.hidden_tag() }}                                                  3621
                <button type="submit" class="btn btn--sm                                 3622
        btn--danger">删除</button>                                                       3623
              </form>                                                                    3624
            </td>                                                                        3625
          </tr>                                                                          3626
          {% endfor %}                                                                   3627
        </tbody>                                                                         3628
      </table>                                                                           3629
      {% else %}                                                                         3630
      <p class="muted">暂无政策文件。</p>                                                3631
      {% endif %}                                                                        3632
</div>                                                                                   3633
{% endblock %}                                                                           3634
# ==== app/templates/standard/edit.html ====                                             3635
{% extends "base.html" %}                                                                3636
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%                          3637
    endblock %}                                                                          3638
{% block content %}                                                                      3639
<div class="card" style="max-width:800px;">                                              3640
  <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>                                  3641
  <form method="post">                                                                   3642
    {{ form.hidden_tag() }}                                                              3643
    <div class="field">                                                                  3644
      <label class="field__label" for="province">省份</label>                            3645
      <input id="province" type="text" name="province"                                   3646
    style="width:160px;" required                                                        3647
            value="{{ standard.province if standard else                                 3648
    '' }}">                                                                              3649
    </div>                                                                               3650
```

```
<div class="field">
  <label class="field__label" for="std-name">名称</label>
  <input id="std-name" type="text" name="name"
style="width:100%;" required
         value="{{ standard.name if standard else ''
}}">
</div>
<div class="field">
  <label class="field__label" for="doc-number">标准编号 /
文号</label>
  <input id="doc-number" type="text" name="doc_number"
style="width:320px;"
         value="{{ standard.doc_number or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="category">类别</label>
  <select id="category" name="category">
    {% for c in ["地方标准", "政策文件"] %}
    <option value="{{ c }}" {% if standard and
standard.category == c %}selected{% endif %}>{{ c
}}</option>
    {% endfor %}
  </select>
</div>
<div class="field">
  <label class="field__label" for="publish-date">发布时间</l
abel>
  <input id="publish-date" type="text"
name="publish_date" style="width:200px;"
         placeholder="如 2023-03-30"
         value="{{ standard.publish_date or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="implement-date">实施时间<
/label>
  <input id="implement-date" type="text"
name="implement_date" style="width:200px;"
         placeholder="如 2023-04-30（未明确可留空）"
         value="{{ standard.implement_date or '' if
standard else '' }}">
</div>
<div class="field">
  <label class="field__label" for="summary">特点 /
判定标准</label>
  <textarea id="summary" name="summary"
style="min-height:96px;">{{ standard.summary or '' if
standard else '' }}</textarea>
</div>
```

```
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        {% if standard %}
        <form method="post" action="/standard/{{ standard.id
      }}/delete" style="display:inline;"
              onsubmit="return confirm('确定删除「{{
      standard.province }} {{ standard.name
      }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn
      btn--danger">删除该标准</button>
        </form>
        {% endif %}
        <a class="btn btn--ghost" href="/standard/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
{% block content %}
<div class="card">
    <h2>结构化规则 · {{ standard.province }} {{ standard.name
      }}</h2>
    <p class="muted">
      将标准条文拆解为「判定项 + 比较符 + 阈值 + 条件类型」的可执行规则，
      使同一指标在各省的阈值差异可自动比对。
    </p>
    {% if rules %}
    <table>
      <caption class="muted">共 {{ rules|length }}
      条规则</caption>
      <thead>
        <tr>
          <th scope="col">判定项</th>
          <th scope="col">条件</th>
          <th scope="col">条件类型</th>
          <th scope="col">适用分区</th>
          <th scope="col">备注</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for r in rules %}
        <tr>
          <td style="text-align:left;">{{ r.item }}</td>
          <td class="mono nowrap">{{ r.operator }} {{
      r.threshold or "" }}</td>
```

```
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
        <td>
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗?');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}
  <h3>添加规则</h3>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="actions">
      <div>
        <label class="field__label" for="r-item">判定项</label>
        <input id="r-item" type="text" name="item" placeholder="如：植被覆盖度" required>
      </div>
      <div>
        <label class="field__label" for="r-op">比较符</label>
        <select id="r-op" name="operator">
          {% for o in [">=", ">", "<=", "<", "=="] %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
        </select>
      </div>
      <div>
        <label class="field__label" for="r-thr">阈值</label>
        <input id="r-thr" type="text" name="threshold" placeholder="如：50" style="width:110px;">
      </div>
      <div>
        <label class="field__label" for="r-cond">条件类型</label>
        <select id="r-cond" name="condition_type">
          <option value="必要条件">必要条件</option>
          <option value="充分条件">充分条件</option>
        </select>
      </div>
```

```
        <div>
          <label class="field__label" for="r-region">适用分区</lab
    el>
          <input id="r-region" type="text" name="region"
    placeholder="如宁夏分区" style="width:130px;">
        </div>
        <div>
          <label class="field__label" for="r-note">备注</label>
          <input id="r-note" type="text" name="note"
    style="width:220px;">
        </div>
        <div>
          <button type="submit" class="btn">添加规则</button>
        </div>
      </div>
    </form>
    <p class="actions" style="margin-top:20px;">
      <a class="btn btn--sm btn--ghost"
      href="/standard/">返回标准列表</a>
      <a class="btn btn--sm btn--ghost"
      href="/standard/compare">省际口径比对</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
    <h2>省际口径比对</h2>
    <p class="muted">
      同一判定项在不同省份的阈值横向对比，避免人工翻阅多份标准文件。
    </p>
    {% if not groups %}
    <p class="muted">
      暂无规则数据。请先在「标准知识库」中为标准录入规则，
      或运行 <span class="mono">python seed_demo.py</span> 载入论文表
      1 数据。
    </p>
    {% else %}
    {% for item, rs in groups.items() %}
    <h3>{{ item }}</h3>
    <table>
      <thead>
        <tr>
          <th scope="col">省份</th>
          <th scope="col">标准 / 文件</th>
          <th scope="col">条件</th>
          <th scope="col">条件类型</th>
          <th scope="col">备注</th>
```

```
        </tr>
      </thead>
      <tbody>
        {% for r in rs %}
        <tr>
          <td>{{ r.standard.province }}</td>
          <td style="text-align:left;">{{ r.standard.name
      }}</td>
          <td class="mono nowrap">{{ r.operator }} {{
      r.threshold or "" }}</td>
          <td>{{ r.condition_type }}</td>
          <td style="text-align:left;">{{ r.note or "—"
      }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endfor %}
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
    <h2>限制性指标判定</h2>
    <p class="muted">
      按论文表 2 与 4.1 节流程，对每项限制性条件逐项判定。
      任一项不通过即判定为「人工辅助或生态重建」，不再进入定量评价。
    </p>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先在「项目与档案」中建立图斑。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">矿山图斑</th>
          <th scope="col">判定结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        {% set r = latest.get(p.id) %}
        <tr>
          <td style="text-align:left;">{{ p.name }}</td>
```

```
            <td style="text-align:left;">
              {% if r %}
                {% if r.passed %}
                <span class="tag tag--ok">通过</span>
                {% else %}
                <span class="tag tag--warn">不通过</span>
                {% endif %}
                <span class="muted">{{ r.reason }}</span>
              {% else %}
              <span class="tag tag--muted">未判定</span>
              {% endif %}
            </td>
            <td><a href="/constraint/{{ p.id }}">执行判定</a></td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
    </div>
    {% endblock %}
    # ==== app/templates/constraint/check.html ====
    {% extends "base.html" %}
    {% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}
    {% block content %}
    <div class="card" style="max-width:760px;">
      <h2>限制性条件判定 · {{ patch.name }}</h2>
      <p class="muted">
        五项条件全部通过方可进入定量评价；任一项不通过即判定为「人工辅助或生态重建」，
        不再进行后续计算。
      </p>
      <form method="post">
        {{ form.hidden_tag() }}
        {% for field, name, val in field_states %}
        <fieldset style="border:1px solid
        var(--line);border-radius:6px;padding:10px
        14px;margin:0 0 10px;">
          <legend style="padding:0 6px;font-weight:600;">{{
        name }}</legend>
          <div class="actions">
            <label class="checkbox">
              <input type="radio" name="{{ field }}" value="1"
                     {% if val is none or val %}checked{% endif
        %}>
              <span>通过</span>
            </label>
            <label class="checkbox">
              <input type="radio" name="{{ field }}" value="0"
                     {% if val is not none and not val
        %}checked{% endif %}>
              <span>不通过</span>
```

```
            </label>                                                    3951
          </div>                                                        3952
        </fieldset>                                                     3953
        {% endfor %}                                                    3954
        <p class="actions" style="margin-top:16px;">                    3955
          <button type="submit" class="btn">保存判定</button>           3956
          <a class="btn btn--ghost" href="/constraint/">返回</a>        3957
        </p>                                                            3958
      </form>                                                           3959
      {% if record %}                                                   3960
      <div style="margin-top:18px;padding:14px;border-radius:6px      3961
        ;border-left:4px solid;                                         3962
                  {% if record.passed %}background:var(--ok-soft       3963
        );border-left-color:var(--ok);                                  3964
                  {% else %}background:var(--danger-soft);border       3965
        -left-color:var(--danger);{% endif %}">                         3966
        <p style="margin:0 0 4px;font-weight:600;">                     3967
          判定结论：                                                    3968
          {% if record.passed %}                                        3969
          <span class="ok">通过，可进入定量评价</span>                  3970
          {% else %}                                                    3971
          <span class="warn">不通过 —— 建议生态重建或人工辅助</span>    3972
          {% endif %}                                                   3973
        </p>                                                            3974
        <p class="muted" style="margin:0;">{{ record.reason             3975
        }}</p>                                                          3976
      </div>                                                            3977
      {% endif %}                                                       3978
</div>                                                                  3979
{% endblock %}                                                          3980
# ==== app/templates/ahp/index.html ====                                3981
{% extends "base.html" %}                                               3982
{% block title %}AHP 权重计算{% endblock %}                             3983
{% block content %}                                                     3984
<div class="card" style="max-width:720px;">                             3985
  <h2>AHP 权重计算 · 选择指标层级</h2>                                  3986
  <p class="muted">                                                     3987
    选择微地质单元类型与指标层级后，系统按论文表 3 确定判断矩阵的规模。 3988
    岩质边坡不设土壤指标、平台不设地形指标，系统会相应调整矩阵所包含的指标。 3989
  </p>                                                                  3990
  <form method="post">                                                  3991
    {{ form.hidden_tag() }}                                             3992
    <div class="field">                                                 3993
      <label class="field__label" for="{{ form.unit_type.id             3994
    }}">微地质单元类型</label>                                          3995
      {{ form.unit_type(id=form.unit_type.id) }}                        3996
    </div>                                                              3997
    <div class="field">                                                 3998
      <label class="field__label" for="{{ form.level.id                 3999
    }}">指标层级</label>                                                4000
```

```
        {{ form.level(id=form.level.id) }}                                  4001
      </div>                                                                4002
      <div class="field">                                                   4003
        <label class="field__label" for="{{ form.l1.id }}">                 4004
          一级指标（层级选「二级指标」时需选择）                                    4005
        </label>                                                            4006
        {{ form.l1(id=form.l1.id) }}                                        4007
      </div>                                                                4008
      <p class="actions" style="margin-top:16px;">                          4009
        {{ form.submit(class_="btn") }}                                     4010
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维护</a>          4011
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a>              4012
      </p>                                                                  4013
    </form>                                                                 4014
</div>                                                                      4015
{% endblock %}                                                              4016
{# ==== app/templates/ahp/matrix.html ==== #}                              4017
{% extends "base.html" %}                                                   4018
{% block title %}判断矩阵录入{% endblock %}                                     4019
{% block content %}                                                         4020
<div class="card">                                                          4021
  <h2>                                                                      4022
    判断矩阵录入                                                                4023
    · {{ unit_type }}                                                       4024
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}          4025
  </h2>                                                                     4026
  <p class="muted">                                                         4027
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，                  4028
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。                 4029
    矩阵已按论文表 3 的权重预填。                                                 4030
  </p>                                                                      4031
  <form method="post">                                                      4032
    {{ form.hidden_tag() }}                                                 4033
    <table style="width:auto;">                                             4034
      <thead>                                                               4035
        <tr>                                                                4036
          <th></th>                                                         4037
          {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}  4038
        </tr>                                                               4039
      </thead>                                                              4040
      <tbody>                                                               4041
        {% for i in range(names|length) %}                                  4042
        <tr>                                                                4043
          <th scope="row">{{ names[i] }}</th>                               4044
          {% for j in range(names|length) %}                                4045
            {% if i == j %}                                                 4046
```

```
                <td class="muted num">1</td>                                    4051
                {% elif i < j %}                                                4052
                <td>                                                            4053
                  <label class="muted" style="position:absolute;                4054
    left:-9999px;"                                                              4055
                         for="m-{{ i }}-{{ j }}">{{ names[i] }}                 4056
    相对 {{ names[j] }} 的重要性</label>                                        4057
                  <input id="m-{{ i }}-{{ j }}" type="number"                   4058
    step="0.01" min="0.01"                                                      4059
                         name="m_{{ i }}_{{ j }}" value="{{                     4060
    default[i][j] }}">                                                          4061
                </td>                                                           4062
                {% else %}                                                      4063
                <td class="muted num">{{ default[i][j] }}</td>                  4064
                {% endif %}                                                     4065
              {% endfor %}                                                      4066
            </tr>                                                               4067
            {% endfor %}                                                        4068
          </tbody>                                                              4069
        </table>                                                                4070
        <p class="actions">                                                     4071
          <button type="submit" class="btn">计算权重</button>                   4072
          <a class="btn btn--ghost" href="/ahp/">重新选择</a>                   4073
        </p>                                                                    4074
    </form>                                                                     4075
    {% if result %}                                                             4076
    <hr style="margin:22px 0;border:none;border-top:1px solid                   4077
      var(--line);">                                                            4078
    <h3>计算结果</h3>                                                           4079
    <table style="width:auto;">                                                 4080
      <thead>                                                                   4081
        <tr>                                                                    4082
          <th scope="col">指标</th>                                             4083
          {% for name in names %}<th scope="col">{{ name                        4084
      }}</th>{% endfor %}                                                       4085
        </tr>                                                                   4086
      </thead>                                                                  4087
      <tbody>                                                                   4088
        <tr>                                                                    4089
          <th scope="row">权重</th>                                             4090
          {% for w in result.weights %}<td class="mono">{{ w                    4091
      }}</td>{% endfor %}                                                       4092
        </tr>                                                                   4093
      </tbody>                                                                  4094
    </table>                                                                    4095
    <div class="stat-row">                                                      4096
      <div>                                                                     4097
        <div class="stat__label">最大特征值 λmax</div>                          4098
        <div class="stat__value mono">{{ result.lambda_max                      4099
      }}</div>                                                                  4100
```

```
      </div>                                                          4101
      <div>                                                           4102
        <div class="stat__label">一致性指标 CI</div>                  4103
        <div class="stat__value mono">{{                             4104
      "%.4f"|format(result.ci) }}</div>                               4105
      </div>                                                          4106
      <div>                                                           4107
        <div class="stat__label">一致性比例 CR</div>                  4108
        <div class="stat__value mono">{{                             4109
      "%.4f"|format(result.cr) }}</div>                               4110
      </div>                                                          4111
      <div>                                                           4112
        <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>  4113
        <div class="stat__value">                                     4114
          {% if result.consistent %}<span class="ok">通过</span>      4115
          {% else %}<span class="warn">不通过</span>{% endif %}       4116
        </div>                                                        4117
      </div>                                                          4118
    </div>                                                            4119
    {% if not result.consistent and result.most_inconsistent          4120
      %}                                                              4121
    <p style="margin-top:14px;padding:12px;border-radius:6px;b        4122
      ackground:var(--danger-soft);                                   4123
                border-left:4px solid var(--danger);">                4124
      <strong class="warn">建议优先调整：</strong>                    4125
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{               4126
      result.most_inconsistent.j + 1 }} 列的比较                      4127
      （当前 <span class="mono num">{{ result.most_inconsistent.v     4128
      alue|round(3) }}</span>，                                       4129
      理想约 <span class="mono num">{{ result.most_inconsistent.e     4130
      xpected|round(3) }}</span>）                                    4131
    </p>                                                              4132
    {% endif %}                                                       4133
    {% endif %}                                                       4134
</div>                                                                4135
{% endblock %}                                                        4136
# ==== app/templates/indicator/index.html ====                        4137
{% extends "base.html" %}                                             4138
{% block title %}指标数据采集{% endblock %}                           4139
{% block content %}                                                   4140
<div class="card">                                                    4141
    <h2>指标数据采集与管理</h2>                                       4142
    <p class="muted">                                                 4143
      按微地质单元组织录入。系统按单元类型自动决定应录入的指标：     4144
      岩质边坡不启用土壤指标、平台不启用地形指标。                   4145
    </p>                                                              4146
    <p class="actions" style="margin-bottom:12px;">                   4147
      <a class="btn btn--sm" href="/indicator/import">批量导入（CSV   4148
      / Excel）</a>                                                   4149
    </p>                                                              4150
```

```
  {% if not patches %}
  <p class="muted">
    暂无矿山图斑。请先在「项目与档案」中建立图斑，
    或运行 <span class="mono">python seed_demo.py</span>
    载入论文案例。
  </p>
  {% else %}
  {% for p in patches %}
  <h3>{{ p.name }}</h3>
  {% if not p.units.count() %}
  <p class="muted">该图斑尚未划分微地质单元，请先到「微地质单元」页面划分。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积占比</th>
        <th scope="col">数据状态</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in p.units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td>
          {% if u.indicator_data %}
          <span class="tag tag--ok">已录入</span>
          {% else %}
          <span class="tag tag--muted">未录入</span>
          {% endif %}
        </td>
        <td><a href="/indicator/{{ u.id }}">录入 / 编辑</a></td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
  {% endfor %}
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/indicator/edit.html ====
{% extends "base.html" %}
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}
{% block content %}
```

```
<div class="card" style="max-width:820px;">
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>
  <p class="muted">
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for group_name, fields, has_hm in groups %}
    <h3>{{ group_name }}</h3>
    {% for fname, cname, unit_ in fields %}
    <div class="field">
      <label class="field__label" for="f-{{ fname }}">
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}
      </label>
      {% if fname == "aspect" %}
      <select id="f-{{ fname }}" name="aspect">
        <option value="">— 不适用 —</option>
        {% for a in aspect_types %}
        <option value="{{ a }}" {% if current.get("aspect")
    == a %}selected{% endif %}>{{ a }}</option>
        {% endfor %}
      </select>
      {% else %}
      <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
             value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
      {% endif %}
    </div>
    {% endfor %}
    {% endfor %}
    {% for group_name, fields, has_hm in groups %}
    {% if has_hm %}
    <h3>重金属含量（mg/kg）</h3>
    <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
    <table style="width:auto;">
      <thead>
        <tr>
          {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        <tr>
          {% for m in heavy_metals %}
          <td>
            <label class="muted" style="position:absolute;
    left:-9999px;"
                   for="hm-{{ m }}">{{ m }} 含量</label>
```

```
                    <input id="hm-{{ m }}" type="number"
      step="any" name="hm_{{ m }}"
                           value="{{ current.get('heavy',
      {}).get(m) if current.get('heavy', {}).get(m) is not
      none else '' }}">
                </td>
                {% endfor %}
              </tr>
            </tbody>
          </table>
          {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% block title %}适宜性评价{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性定量评价</h2>
    <p class="muted">
      系统按论文式（8） 对各项指标做标准化，按式（9） 分单元赋分并按面积占比加权合成，
      最后按论文表 4 判定适宜性等级。
    </p>
    {% if not patches %}
    <p class="muted">暂无矿山图斑。请先在「项目与档案」中建立图斑并录入指标数据。</p>
    {% else %}
    <table>
      <thead>
        <tr>
          <th scope="col">矿山图斑</th>
          <th scope="col">最近评价结果</th>
          <th scope="col">操作</th>
        </tr>
      </thead>
      <tbody>
        {% for p in patches %}
        <tr>
          <td style="text-align:left;">{{ p.name }}</td>
          <td>
            {% if latest.get(p.id) %}
            F = <span class="mono num">{{
      latest[p.id].final_score }}</span>,
            <span class="tag tag--ok">{{ latest[p.id].grade
      }}</span>
```

```
            {% else %}
            <span class="tag tag--muted">未评价</span>
            {% endif %}
          </td>
          <td class="nowrap">
            <a href="/evaluate/{{ p.id }}">执行评价</a>
            {% if latest.get(p.id) %}
             | <a href="/evaluate/result/{{
    latest[p.id].id }}">查看结果</a>
            {% endif %}
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/evaluate/run.html ====
{% extends "base.html" %}
{% block title %}执行评价 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:640px;">
    <h2>执行适宜性评价 · {{ patch.name }}</h2>
    <p class="muted">
      系统将对图斑内各微地质单元做指标标准化、分单元赋分、面积加权合成，
      并判定适宜性等级。评价结果会保存，可随时回看。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">开始评价</button>
        <a class="btn btn--ghost" href="/evaluate/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/evaluate/result.html ====
{% extends "base.html" %}
{% block title %}评价结果{% endblock %}
{% block content %}
<div class="card">
    <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">最终评价因子 F</div>
        <div class="stat__value mono">{{ r.final_score
      }}</div>
      </div>
      <div>
```

```
        <div class="stat__label">适宜性等级</div>                                4351
        <div class="stat__value ok">{{ r.grade }}</div>                          4352
      </div>                                                                     4353
      <div style="flex:1;min-width:240px;">                                      4354
        <div class="stat__label">推荐措施</div>                                  4355
        <div style="font-size:15px;">{{ r.measures }}</div>                      4356
      </div>                                                                     4357
    </div>                                                                       4358
    {% if unit_scores %}                                                         4359
    <h3>各微地质单元得分</h3>                                                    4360
    <table style="max-width:720px;">                                             4361
      <thead>                                                                    4362
        <tr>                                                                     4363
          <th scope="col">微地质单元</th>                                        4364
          <th scope="col">面积占比</th>                                          4365
          <th scope="col">单元评价因子 F'</th>                                   4366
          <th scope="col">加权贡献</th>                                          4367
        </tr>                                                                    4368
      </thead>                                                                   4369
      <tbody>                                                                    4370
        {% for u in unit_scores %}                                               4371
        <tr>                                                                     4372
          <td>{{ u.unit_type }}</td>                                             4373
          <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                      4374
      }}</td>                                                                    4375
          <td class="mono">{{ u.unit_score }}</td>                               4376
          <td class="mono">{{ "%.4f"|format(u.ratio *                            4377
      u.unit_score) }}</td>                                                      4378
        </tr>                                                                    4379
        {% endfor %}                                                             4380
      </tbody>                                                                   4381
    </table>                                                                     4382
    {% endif %}                                                                  4383
    <p class="actions" style="margin-top:18px;">                                 4384
      <a class="btn btn--ghost" href="/report/{{ r.patch_id                      4385
      }}">查看得分明细</a>                                                       4386
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>               4387
    </p>                                                                         4388
</div>                                                                           4389
{% endblock %}                                                                   4390
# ==== app/templates/report/index.html ====                                      4391
{% extends "base.html" %}                                                        4392
{% block title %}报告输出{% endblock %}                                          4393
{% block content %}                                                              4394
<div class="card">                                                               4395
  <h2>报告与结果输出</h2>                                                        4396
  <p class="muted">可查看逐层得分明细，并导出规范的评价报告文档。</p>           4397
  {% if not patches %}                                                           4398
  <p class="muted">暂无矿山图斑。请先在「项目与档案」中建立图斑并完成评价。</p> 4399
  {% else %}                                                                     4400
```

```
  <table>                                                                    4401
    <thead>                                                                  4402
      <tr>                                                                   4403
        <th scope="col">矿山图斑</th>                                        4404
        <th scope="col">操作</th>                                            4405
      </tr>                                                                  4406
    </thead>                                                                 4407
    <tbody>                                                                  4408
      {% for p in patches %}                                                 4409
      <tr>                                                                   4410
        <td style="text-align:left;">{{ p.name }}</td>                       4411
        <td class="nowrap">                                                  4412
          <a href="/report/{{ p.id }}">查看得分明细</a>                      4413
           | <a href="/report/{{ p.id }}/pdf">导出                 4414
    PDF</a>                                                                  4415
           | <a href="/report/{{ p.id }}/word">导出                4416
    Word</a>                                                                 4417
        </td>                                                                4418
      </tr>                                                                  4419
      {% endfor %}                                                           4420
    </tbody>                                                                 4421
  </table>                                                                   4422
  {% endif %}                                                                4423
</div>                                                                       4424
{% endblock %}                                                               4425
# ==== app/templates/report/detail.html ====                                 4426
{% extends "base.html" %}                                                    4427
{% block title %}得分明细 · {{ patch.name }}{% endblock %}                   4428
{% block content %}                                                          4429
<div class="card">                                                           4430
  <h2>得分明细 · {{ patch.name }}</h2>                                       4431
  <div class="stat-row">                                                     4432
    <div>                                                                    4433
      <div class="stat__label">最终评价因子 F</div>                          4434
      <div class="stat__value mono">{{                                       4435
    "%.4f"|format(result.final_score) }}</div>                               4436
    </div>                                                                   4437
    <div>                                                                    4438
      <div class="stat__label">适宜性等级</div>                              4439
      <div class="stat__value ok">{{ result.grade }}</div>                   4440
    </div>                                                                   4441
    <div style="flex:1;min-width:240px;">                                    4442
      <div class="stat__label">推荐措施</div>                                4443
      <div style="font-size:15px;">{{                                        4444
    result.grade_info.measure }}</div>                                       4445
    </div>                                                                   4446
  </div>                                                                     4447
  <p class="actions" style="margin-top:16px;">                               4448
    <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF                 4449
    报告</a>                                                                 4450
```

```
      <a class="btn btn--ghost" href="/report/{{ patch.id }}/word">导出 Word 报告</a>
      <a class="btn btn--ghost" href="/report/">返回</a>
    </p>
</div>
<div class="card">
    <h2>结果图表</h2>
    <h3>各微地质单元得分对比</h3>
    <div>{{ chart_units|safe }}</div>
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
    <div>{{ chart_weight|safe }}</div>
</div>
{% for u in result.unit_results %}
<div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio * 100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td style="text-align:left;">{{ d.l2 }}</td>
          <td class="mono" style="text-align:left;">{{ "—" if d.raw is none else d.raw }}</td>
          <td class="mono">{{ "—" if d.std_value is none else "%.4f"|format(d.std_value) }}</td>
          <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
          <td class="mono">{{ "—" if d.contribution is none else "%.4f"|format(d.contribution) }}</td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="muted" style="margin:0;">
      一级指标得分：
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span class="num">{{ "%.4f"|format(v) }}</span>{% if not loop.last %}；{% endif %}{% endfor %}
    </p>
```

```
</div>
{% endfor %}
{% endblock %}
# ==== seed_demo.py ====
# -*- coding: utf-8 -*-
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。
运行：python seed_demo.py（幂等：已存在则先清空重建）
"""
import json
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
from app.models import (db, User, Project, MinePatch,
    ReferenceEcosystem,
                        MicroUnit, IndicatorData, Standard,
    StandardRule)
app = create_app()
# 论文表 1：11 项地方标准与政策文件
STANDARDS = [
    dict(category="地方标准", province="辽宁",
    name="矿山废弃地植被自然恢复技术规程",
        doc_number="DB21/T 3702—2023",
    publish_date="2023-03-30",
        implement_date="2023-04-30",
        summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",
        rules=[("植被覆盖度", ">=", "50", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
    dict(category="地方标准", province="湖北",
    name="废弃露天矿山生态环境自然恢复标准",
        doc_number="DB42/T 2421—2025",
    publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="内蒙古",
    name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
    publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
    dict(category="地方标准", province="江西",
    name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
```

```
    publish_date="２０２６-０２-０２",                                    4551
        implement_date="２０２６-０８-０１",                             4552
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",      4553
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),       4554
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),  4555
dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",  4556
        doc_number=None, publish_date="２０２６-０６-１８",              4557
implement_date=None,                                                    4558
        summary="分区差异化：三个一级生态修复分区及 ２２ 个二级分区",    4559
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),         4560
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),4561
               ("与参照系差距", "<", "２０", "充分条件", "%")]),         4562
dict(category="政策文件", province="安徽",                              4563
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",4564
        doc_number="皖自然资修函[２０２０]３４号",                       4565
publish_date="２０２０-０３-０３", implement_date=None,                 4566
        summary="自然恢复条件：责任主体为政府、无地质灾害、不在三线三边、水土条件良好",4567
        rules=[("责任主体", "==", "政府", "必要条件", None),             4568
               ("无地质灾害", "==", "是", "必要条件", None),             4569
               ("无水土污染", "==", "是", "必要条件", None)]),           4570
dict(category="政策文件", province="山东",                              4571
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",4572
        doc_number="鲁自然资字[２０２２]１０４号",                       4573
publish_date="２０２２-０８-０２", implement_date=None,                 4574
        summary="条件与安徽省相同，另增加自然恢复认定标准",             4575
        rules=[("无地质灾害", "==", "是", "必要条件", None),             4576
               ("无水土污染", "==", "是", "必要条件", None)]),           4577
dict(category="政策文件", province="湖南",                              4578
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",4579
        doc_number="湘自资办发[２０２３]５８号",                         4580
publish_date="２０２３-０７-２０", implement_date=None,                 4581
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度4582
＞７５%、连续３年正向增长",                                             4583
        rules=[("植被覆盖度", ">", "７５", "必要条件", "%"),             4584
               ("植被覆盖率趋势", "==", "连续３年正向增长", "必要条件", None),4585
               ("无地质灾害", "==", "是", "必要条件", None),             4586
               ("符合国土空间规划", "==", "是", "必要条件", None)]),     4587
dict(category="政策文件", province="新疆",                              4588
        name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",4589
        doc_number="新自然资源办函〔２０２４〕２１９号",                 4590
publish_date="２０２４-１２-２０", implement_date=None,                 4591
        summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",4592
        rules=[("植被覆盖度", ">=", "周边３０%", "必要条件", None),      4593
               ("无地质灾害", "==", "是", "必要条件", None),             4594
               ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),     4595
dict(category="政策文件", province="广西",                              4596
        name="广西壮族自治区历史遗留矿山自然恢复技术指南",              4597
        doc_number="桂自然资办〔２０２４〕２５０号",                     4598
publish_date="２０２４-１２-３０", implement_date=None,                 4599
        summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",4600
```

```
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
    publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
                     "slope": 28.75, "aspect": "阳坡"},
        },
        {
            "unit_type": "平台", "ratio": 0.2319,
            "data": {"soil_ratio": 0.56, "thickness": 30,
    "ph": 8.72,
                     "organic_matter": 10.92,
                     "heavy_metals": {"Hg": 0.02, "As":
    3.13, "Ni": 12.33,
                                      "Cu": 11.14, "Cd":
    0.09, "Pb": 9.98},
                     "coverage": 58.42, "shannon": 0.72},
        },
    ],
}
def clean_case():
    """清空案例相关数据（幂等重跑用）。
```

```
    indicator_data / score_details 等表未设级联删除，需显式清理。
    """
    from app.models import EvaluationResult, ScoreDetail,
    ConstraintCheck
    ScoreDetail.query.delete()
    EvaluationResult.query.delete()
    ConstraintCheck.query.delete()
    IndicatorData.query.delete()
    MicroUnit.query.delete()
    ReferenceEcosystem.query.delete()
    MinePatch.query.delete()
    Project.query.delete()
    db.session.commit()
def seed():
    with app.app_context():
        db.create_all()        # 全新环境下先建表，保证本脚本可独立运行
        clean_case()
        admin = User.query.filter_by(username="admin").first
    ()
        if not admin:
            from werkzeug.security import
    generate_password_hash
            admin = User(username="admin",
                         password_hash=generate_password_has
    h("admin123"),
                         role="admin")
            db.session.add(admin)
            db.session.commit()
        project = Project(name=CASE["name"],
    code="DEMO-001",
                          description="论文第 5 节典型案例",
    created_by=admin.id)
        db.session.add(project)
        db.session.flush()
        patch = MinePatch(project_id=project.id,
    name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
    stop_years=None,
                          responsible_entity_missing=True)
        db.session.add(patch)
        db.session.flush()
        ref = ReferenceEcosystem(patch_id=patch.id,
                                 coverage_benchmark=75.0,
    shannon_benchmark=0.7)
        db.session.add(ref)
        for u in CASE["units"]:
            unit = MicroUnit(patch_id=patch.id,
```

```
        unit_type=u["unit_type"],                                          4701
                                    ratio=u["ratio"], area=None)           4702
                db.session.add(unit)                                       4703
                db.session.flush()                                         4704
                d = u["data"]                                              4705
                ind = IndicatorData(                                       4706
                    unit_id=unit.id,                                       4707
                    soil_ratio=d.get("soil_ratio"),                        4708
                    thickness=d.get("thickness"),                          4709
                    ph=d.get("ph"),                                        4710
                    organic_matter=d.get("organic_matter"),                4711
                    heavy_metals=json.dumps(d["heavy_metals"],             4712
        ensure_ascii=False)                                                4713
                                      if d.get("heavy_metals") else        4714
        None,                                                              4715
                    coverage=d.get("coverage"),                            4716
                    shannon=d.get("shannon"),                              4717
                    slope=d.get("slope"),                                  4718
                    aspect=d.get("aspect"),                                4719
                )                                                          4720
                db.session.add(ind)                                        4721
            db.session.commit()                                            4722
            print("已写入案例：", CASE["name"])                              4723
            print("  图斑：", CASE["patch_name"])                           4724
            print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台      4725
        23.19%")                                                           4726
            print("  管理员账号：admin / admin123")                         4727
            seed_standards()                                               4728
            # 指标定义（M6 指标体系维护）                                    4729
            from app.services.indicator_service import                     4730
        seed_indicators                                                    4731
            n = seed_indicators()                                          4732
            print("已写入指标定义：%d 项" % n)                               4733
    def seed_standards():                                                  4734
        """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""              4735
        StandardRule.query.delete()                                        4736
        Standard.query.delete()                                            4737
        db.session.commit()                                                4738
        for s in STANDARDS:                                                4739
            std = Standard(category=s["category"],                         4740
        province=s["province"],                                            4741
                           name=s["name"],                                 4742
        doc_number=s["doc_number"],                                        4743
                           publish_date=s["publish_date"],                 4744
                           implement_date=s["implement_date"],             4745
                           summary=s["summary"])                           4746
            db.session.add(std)                                            4747
            db.session.flush()                                             4748
            for item, op, thr, ctype, note in s["rules"]:                  4749
                db.session.add(StandardRule(standard_id=std.id,            4750
```

```
        item=item,                                                    4751
                                                  operator=op,        4752
        threshold=thr,                                                4753
                                                                      4754
        condition_type=ctype, note=note))                             4755
        db.session.commit()                                           4756
        print("已写入标准/政策文件：%d 项，规则 %d 条"                    4757
              % (len(STANDARDS), sum(len(s["rules"]) for s in         4758
        STANDARDS)))                                                  4759
if __name__ == "__main__":                                            4760
        seed()                                                        4761
#  ==== test_ahp.py ====                                             4762
#  -*- coding: utf-8 -*-                                             4763
"""地基自检：验证 AHP 引擎与应用可启动。"""                              4764
import sys                                                            4765
sys.stdout.reconfigure(encoding="utf-8")                              4766
import numpy as np                                                    4767
from app.services import ahp_service as ahp                           4768
LINE = "=" * 66                                                       4769
#  ---------------------------------------------------------------------4770
        ------------------                                            4771
#  1） 完全一致矩阵 → 权重应精确还原，CR=0                              4772
#  ---------------------------------------------------------------------4773
        ------------------                                            4774
print(LINE)                                                           4775
print("测试 1  完全一致判断矩阵的权重还原与一致性")                      4776
print(LINE)                                                           4777
#  由权重构造完全一致矩阵：a_ij = w_i / w_j                           4778
w_true = np.array([0.44, 0.31, 0.25])                                 4779
A = np.outer(w_true, 1.0 / w_true)                                    4780
r = ahp.weights_and_consistency(A)                                    4781
print("  判断矩阵 A：")                                              4782
for row in A:                                                         4783
        print("      " + "  ".join(f"{v:8.4f}" for v in row))         4784
print()                                                               4785
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in                   4786
        r["weights"]))                                                4787
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))          4788
print("  λmax = %.6f    CI = %.3e    CR = %.3e    一致: %s"            4789
        % (r["lambda_max"], r["ci"], r["cr"],                         4790
        r["consistent"]))                                             4791
assert np.allclose(r["weights"], w_true, atol=1e-6),                  4792
        "权重未精确还原"                                              4793
assert r["consistent"], "完全一致矩阵应通过检验"                       4794
#  ---------------------------------------------------------------------4795
        ------------------                                            4796
#  2） 上三角输入 → 完整矩阵 → 一致性定位                              4797
#  ---------------------------------------------------------------------4798
        ------------------                                            4799
print()                                                               4800
```

```
print(LINE)                                                                    4801
print("测试 2  上三角录入与最不一致项定位")                                    4802
print(LINE)                                                                    4803
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）             4804
M = ahp.matrix_from_upper([2, 5, 3])                                           4805
r2 = ahp.weights_and_consistency(M)                                            4806
bad = ahp.find_most_inconsistent(M)                                            4807
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")                                4808
for row in M:                                                                  4809
    print("    " + "  ".join(f"{v:8.4f}" for v in row))                        4810
print("  权重：", "  ".join(f"{v:.4f}" for v in r2["weights"]),                4811
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))                  4812
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"                   4813
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],                             4814
         bad["expected"]))                                                     4815
# ------------------------------------------------------------------------     4816
    ----------                                                                 4817
# 3） 多专家聚合（几何平均）                                                   4818
# ------------------------------------------------------------------------     4819
    ----------                                                                 4820
print()                                                                        4821
print(LINE)                                                                    4822
print("测试 3  多专家判断矩阵聚合（几何平均）")                                4823
print(LINE)                                                                    4824
m1 = ahp.matrix_from_upper([2, 4, 3])                                          4825
m2 = ahp.matrix_from_upper([3, 5, 2])                                          4826
agg = ahp.aggregate_matrices([m1, m2])                                         4827
print("  专家1 上三角 [2,4,3]，专家2 上三角 [3,5,2]")                         4828
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"                            4829
      % (agg[0, 1], agg[0, 2], agg[1, 2]))                                     4830
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"                  4831
# ------------------------------------------------------------------------     4832
    ----------                                                                 4833
# 4） 论文表 3 权重可还原性                                                    4834
# ------------------------------------------------------------------------     4835
    ----------                                                                 4836
print()                                                                        4837
print(LINE)                                                                    4838
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")                  4839
print(LINE)                                                                    4840
j = ahp.judge_from_weights([0.44, 0.31, 0.25])                                 4841
print("  λmax = %.6f  CR = %.3e  一致: %s" %                                   4842
    (j["lambda_max"], j["cr"], j["consistent"]))                               4843
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"               4844
# ------------------------------------------------------------------------     4845
    ----------                                                                 4846
# 5） 应用可启动 + 建表                                                        4847
# ------------------------------------------------------------------------     4848
    ----------                                                                 4849
print()                                                                        4850
```

```
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表：" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检：用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
    ------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------------
    ------------------
# 三类单元面积占比：岩质边坡 13.86%、碎石边坡 62.95%、平台 23.19%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
```

```
                "coverage": 87.92, "shannon": 0.60,
                "slope": 28.75, "aspect": "阳坡",
            },
        },
        {
            "unit_type": "平台",
            "ratio": 0.2319,
            "data": {
                "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
                "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                                 "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
                "coverage": 58.42, "shannon": 0.72,
            },
        },
]
# ------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
```

```
print("   判定依据   ：%s" % res["grade_info"]["meaning"])
print("   推荐措施   ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("   - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("   - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("   - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("   - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("     该口径在 standardization.std_slope / aspect_score 中集中定义、可配置")
print(LINE)
# ==== test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""M6 AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("   首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
```

```
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」：", "计算结果" in body)
print("  含权重 0.44：", "0.44" in body, " 含 0.31：", "0.31" in
    body, " 含 0.25：", "0.25" in body)
print("  含一致性检验通过：", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」：", "不通过" in body)
print("  含「建议优先调整」：", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("M6 网页界面全部测试通过 √")
print(LINE)
```

```
# ==== test_web_modules.py ====
# -*- coding: utf-8 -*-
"""M1–M5 模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
print("  登录后含「系统功能」:", "系统功能" in body)
print("  含用户名 admin:", "admin" in body)
assert "系统功能" in body
print()
print(LINE)
print("测试 3  M2 项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
```

```
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  M5 微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元：", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("  含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5  M3 标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("  含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  M4 限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件：", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
```

```
        "policy_pass": "1", "location_pass": "1",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "0",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
print()
print(LINE)
print("M1–M5 全部测试通过 √")
print(LINE)
# ==== test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""M7 + M8 端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
        SECRET_KEY = "test"
        SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
                os.path.dirname(__file__), "app", "app.db")
        SQLALCHEMY_TRACK_MODIFICATIONS = False
        WTF_CSRF_ENABLED = False
        TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
        "password": "admin123"})
print(LINE)
print("测试 1  M7 指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码:", r.status_code)
print("  含案例图斑:", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元:", all(n in body for n in ("岩质边坡", "碎石边坡",
```

```
        "平台"))) 5201
assert r.status_code == 200 5202
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body 5203
print() 5204
print(LINE) 5205
print("测试 2  M7 指标数据录入页（按单元类型显示字段）") 5206
print(LINE) 5207
from app.models import MicroUnit, MinePatch 5208
with app.app_context(): 5209
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").5210
    first() 5211
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"5212
    ).first() 5213
r = client.get("/indicator/%d" % unit_rock.id) 5214
b_rock = r.get_data(as_text=True) 5215
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）") 5216
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock) 5217
assert "土石比" not in b_rock and "植被覆盖度" in b_rock 5218
r = client.get("/indicator/%d" % unit_gravel.id) 5219
b_gravel = r.get_data(as_text=True) 5220
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel) 5221
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in 5222
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))) 5223
assert "土石比" in b_gravel and all(m in b_gravel for m in 5224
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")) 5225
print() 5226
print(LINE) 5227
print("测试 3  M8 执行评价") 5228
print(LINE) 5229
with app.app_context(): 5230
    patch = MinePatch.query.first() 5231
r = client.post("/evaluate/%d" % patch.id, data={}, 5232
    follow_redirects=True) 5233
body = r.get_data(as_text=True) 5234
print("  跳转至结果页：", "适宜性评价结果" in body) 5235
print("  含最终评价因子 F：", "最终评价因子 F" in body) 5236
print("  含等级「适宜」：", "适宜" in body) 5237
print("  含推荐措施「自然恢复」：", "自然恢复" in body) 5238
assert "最终评价因子 F" in body and "自然恢复" in body 5239
print() 5240
print(LINE) 5241
print("测试 4  结果持久化") 5242
print(LINE) 5243
from app.models import EvaluationResult 5244
with app.app_context(): 5245
    res = EvaluationResult.query.order_by(EvaluationResult.i5246
    d.desc()).first() 5247
    print("  最终得分 F =", res.final_score) 5248
    print("  等级 =", res.grade) 5249
    assert res.grade == "适宜" 5250
```

```
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("M7 + M8 端到端全部测试通过 √")
print(LINE)
# ==== test_web_report.py ====
# -*- coding: utf-8 -*-
"""M9 报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.first()
    pid = patch.id
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码:", r.status_code)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码:", r.status_code)
print("  含「得分明细」:", "得分明细" in body)
print("  含「最终评价因子 F」:", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）:", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列:", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
```

```
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码:", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小:", len(data), "字节")
print("  是否为有效 docx(PK 头):", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题:", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节:", "评价结论" in text)
print("  含得分明细章节:", "得分明细" in text)
print("  段落数:", len(d.paragraphs), " 表格数:", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("M9 报告输出全部测试通过 √")
print(LINE)
# ==== test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检:「记住我」与各模块删除功能。"""
import os
import re
import sys
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、「记住我」")
print(LINE)
```

```
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
wrapped = re.search(r'<label[^>]*class="checkbox"[^>]*>\s*<input[^>]*name="remember"',
                    html) is not None
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password": "admin123",
                           "remember": "y"}, follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin", "password": "admin123"},
             follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2 else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1], domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]", r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------------------
print()
```

```
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表
with app.app_context():
    from app.models import MinePatch, Project
    patch = MinePatch.query.first()
    patch_id = patch.id
html = client.get("/project/%d/patches" %
    patch_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
```

```
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：M1 操作日志与用户维护、M2 遥感影像、M6 指标与多专家矩阵、
M7 批量导入、M9 图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = os.path.dirname(os.path.abspath(__file__))
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    patch_id = MinePatch.query.first().id
c = app.test_client()
```

```
c.post("/login", data={"username": "admin", "password": "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]", label, extra))
    assert flag, label
# ------------------------------------------------------------------------
print(LINE)
print("M1 操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code": "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc()).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before, after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型", last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator", "active": "1"})
with app.app_context():
    ok(User.query.get(uid).role == "evaluator", "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer", "active": "1"})
with app.app_context():
    ok(User.query.get(admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
```

```
with app.app_context():
    ok(User.query.get(uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("M2 遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(png), "影像.png"),
                  "image_note": "2025年6月无人机正射影像"},
            content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = MinePatch.query.get(patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
            data={"image": (io.BytesIO(b"x"), "bad.txt")},
            content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(MinePatch.query.get(patch_id).image_path is None,
    "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ----------------------------------------------------------------
    ----------------
print()
print(LINE)
print("M6 指标定义维护与多专家矩阵")
```

```
print(LINE)                                                                    5601
r = c.get("/ahp/indicators")                                                   5602
body = r.get_data(as_text=True)                                                5603
with app.app_context():                                                        5604
    n_ind = Indicator.query.count()                                            5605
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d          5606
    项" % n_ind)                                                               5607
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),               5608
   "指标树按层级展示")                                                         5609
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")        5610
c.post("/ahp/indicators/coverage/edit",                                        5611
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":                  5612
    "benefit",                                                                 5613
              "ideal_value": "75", "enabled_units": ["岩质边坡"],              5614
              "description": "自检修改"})                                      5615
with app.app_context():                                                        5616
    ok(Indicator.query.filter_by(code="coverage").first().de                   5617
    scription == "自检修改",                                                   5618
        "指标定义可编辑")                                                      5619
c.post("/ahp/indicators/reset")                                                5620
with app.app_context():                                                        5621
    ok("自检修改" not in (Indicator.query.filter_by(code="covera               5622
    ge").first().description or ""),                                           5623
        "可按论文取值重置")                                                    5624
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}                   5625
for name in ("张工", "李工", "王工"):                                          5626
    d = dict(base); d.update({"action": "add", "unit_type":                    5627
    "碎石边坡",                                                                5628
                              "level": "1", "expert_name":                     5629
    name})                                                                     5630
    c.post("/ahp/experts", data=d)                                             5631
with app.app_context():                                                        5632
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",                      5633
    level=1).count() == 3,                                                     5634
        "录入 3 位专家的判断矩阵")                                             5635
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")                 5636
body = r.get_data(as_text=True)                                                5637
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,                      5638
   "多专家矩阵几何平均汇总可用")                                               5639
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",                    5640
    body, re.S)                                                                5641
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws                       5642
    else []                                                                    5643
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <                       5644
    0.01,                                                                      5645
   "汇总权重归一", "、".join(vals))                                            5646
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",             5647
    vals[0] if vals else "")                                                   5648
# ------------------------------------------------------------------------     5649
    ----------------                                                           5650
```

```
print()
print(LINE)
print("M7 批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,58.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("utf-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first()
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("M9 图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
```

```
        "0.4000")])                                                    5701
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                5702
r = c.get("/report/%d" % patch_id)                                     5703
body = r.get_data(as_text=True)                                        5704
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %           5705
        body.count("<svg"))                                            5706
exe = browser.browser_path()                                           5707
if exe is None:                                                        5708
        print("  [跳过] 未找到浏览器，PDF 导出无法验证")                  5709
else:                                                                  5710
        r = c.get("/report/%d/pdf" % patch_id)                         5711
        data = r.data                                                  5712
        ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %        5713
        len(data))                                                     5714
        d = fitz.open(stream=data, filetype="pdf")                     5715
        txt = "".join(d[i].get_text() for i in                         5716
        range(d.page_count))                                           5717
        ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")             5718
        ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")  5719
        ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)  5720
        d.close()                                                      5721
print()                                                                5722
print(LINE)                                                            5723
print("补齐模块自检全部通过 √")                                         5724
print(LINE)                                                            5725
# ==== app/services/__init__.py ====                                   5726
# -*- coding: utf-8 -*-                                                5727
# ==== app/services/audit.py ====                                      5728
# -*- coding: utf-8 -*-                                                5729
"""操作审计（M1）。                                                     5730
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）  5731
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。           5732
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。   5733
"""                                                                    5734
from flask import request                                              5735
from flask_login import current_user                                   5736
from app.models import db, OperationLog                                5737
# 端点 → 可读动作名                                                    5738
ACTION_NAMES = {                                                       5739
        "auth.login": "登录系统",                                      5740
        "auth.logout": "退出登录",                                     5741
        "auth.add_user": "新增用户",                                   5742
        "auth.edit_user": "编辑用户",                                  5743
        "auth.delete_user": "删除用户",                                5744
        "auth.reset_password": "重置用户密码",                         5745
        "project.add": "新建项目",                                     5746
        "project.edit": "编辑项目",                                    5747
        "project.delete": "删除项目",                                  5748
        "project.patch_add": "新建矿山图斑",                           5749
        "project.patch_edit": "编辑矿山图斑",                          5750
```

```
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
}
# 无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
```

```
                target_id=extract_target_id(),
                detail=detail,
            ))
            db.session.commit()
        except Exception:
            # 审计失败不应影响业务响应
            db.session.rollback()
# ==== app/services/import_service.py ====
# -*- coding: utf-8 -*-
"""指标数据批量导入（M7）。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg, As, Ni, Cu, Cd, Pb, Cr, Zn,
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
# 表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    # 示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
```

```
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
```

```
    返回                                                                    5901
    ----                                                                    5902
    (parsed, errors)                                                        5903
       parsed : list[dict]，每项 {unit_type, fields{...},                   5904
    heavy_metals{}}                                                         5905
       errors : list[str]，逐行问题说明                                     5906
    """                                                                     5907
    rows = _read_rows(filename, stream)                                     5908
    parsed, errors = [], []                                                 5909
    if not rows:                                                            5910
        return [], ["文件为空或没有数据行"]                                 5911
    # 表头校验                                                              5912
    unknown = [h for h in (rows[0].keys() if rows else [])                  5913
    if h not in FIELD_ALIASES]                                              5914
    if unknown:                                                             5915
        errors.append("存在无法识别的列，将被忽略：%s" %                    5916
    "、".join(map(str, unknown)))                                           5917
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头        5918
        rec = {}                                                            5919
        for k, v in raw.items():                                            5920
            key = FIELD_ALIASES.get(str(k).strip())                         5921
            if key:                                                         5922
                rec[key] = v                                                5923
        unit_type = str(rec.get("unit_type") or "").strip()                 5924
        if not unit_type:                                                   5925
            errors.append("第 %d 行：缺少「单元类型」" % idx)               5926
            continue                                                        5927
        if unit_type not in unit_type_map:                                  5928
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"  5929
                          % (idx, unit_type,                                5930
    "、".join(unit_type_map) or "无"))                                      5931
            continue                                                        5932
        if unit_type not in UNIT_TYPES:                                     5933
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"   5934
                          % (idx, unit_type))                               5935
            continue                                                        5936
        fields, heavy = {}, {}                                              5937
        ok = True                                                           5938
        for key, val in rec.items():                                        5939
            if key in ("unit_type",):                                       5940
                continue                                                    5941
            if key.startswith("hm_"):                                       5942
                n, err = _num(val)                                          5943
                if err:                                                     5944
                    errors.append("第 %d 行：%s %s" % (idx,                 5945
    key[3:], err))                                                          5946
                    ok = False                                              5947
                elif n is not None:                                         5948
                    heavy[key[3:]] = n                                      5949
                continue                                                    5950
```

```
                if key == "aspect":
                    s = str(val).strip() if val is not None
    else ""
                    if s == "":
                        fields["aspect"] = None
                    elif s in ASPECT_TYPES:
                        fields["aspect"] = s
                    else:
                        errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                      % (idx, s,
    "、".join(ASPECT_TYPES)))
                        ok = False
                    continue
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx, key,
    err))
                    ok = False
                else:
                    fields[key] = n
            if ok:
                parsed.append({"unit_type": unit_type,
    "fields": fields,
                               "heavy_metals": heavy})
    return parsed, errors
# ==== app/services/indicator_service.py ====
# -*- coding: utf-8 -*-
"""指标定义维护（M6）。
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。
"""
from app.models import db, Indicator
from app.utils.constants import (
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,
    IDEAL_THICKNESS,
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,
    IDEAL_SHANNON,
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,
)
ALL_UNITS = ",".join(UNIT_TYPES)
# (code, name, level, parent, unit, ideal_value,
    ideal_type, enabled_units, desc, is_constraint)
TREE = [
    # ---- 一级指标 ----
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",
    "论文表 2 评价指标之一级指标", False),
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,
    "论文表 2 评价指标之一级指标", False),
```

```
        ("terrain", "地形指标", 1, None, None, None, None,                6001
        "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),                6002
        # ---- 二级指标：土壤 ----                                          6003
        ("soil_ratio", "土石比", 2, "soil", "—",                          6004
        str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",                 6005
         "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),          6006
        ("thickness", "土层厚度", 2, "soil", "cm",                        6007
        str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",                  6008
         "论文 4.2.2 节：理想值 20 cm", False),                           6009
        ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,            6010
        "mid", "碎石边坡,平台",                                            6011
         "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %              6012
        IDEAL_PH_RANGE, False),                                           6013
        ("organic_matter", "有机质", 2, "soil", "g/kg",                   6014
        str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",                    6015
         "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),  6016
        ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",          6017
        "cost", "碎石边坡,平台",                                           6018
         "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),     6019
        # ---- 二级指标：植被 ----                                          6020
        ("coverage", "植被覆盖度", 2, "veg", "%",                         6021
        str(IDEAL_COVERAGE), "benefit", ALL_UNITS,                        6022
         "论文 4.2.2 节：参照生态系统取值 75%", False),                       6023
        ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),       6024
        "benefit", ALL_UNITS,                                             6025
         "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),           6026
        # ---- 二级指标：地形 ----                                          6027
        ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %                 6028
        SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",                          6029
         "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°   6030
        为陡坡重要分界", False),                                           6031
        ("aspect", "坡向", 2, "terrain", "—",                             6032
        "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",       6033
         "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),          6034
]                                                                         6035
# 污染指标下的三级指标（8 种重金属）                                          6036
for _m, _limit in HEAVY_METAL_LIMIT.items():                              6037
    TREE.append((                                                         6038
        "hm_" + _m, _m, 3, "pollution", "mg/kg",                          6039
    str(_limit), "cost",                                                  6040
        "碎石边坡,平台",                                                   6041
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %      6042
    _limit,                                                               6043
        False,                                                            6044
    ))                                                                    6045
def seed_indicators():                                                    6046
    """写入/更新指标定义（幂等）。返回写入条数。"""                                6047
    code_map = {                                                          6048
        "benefit": "越大越优", "cost": "越小越优",                         6049
        "mid": "适中型", "discrete": "离散赋值",                           6050
```

```
    }                                                                    6051
    n = 0                                                                6052
    for (code, name, level, parent, unit, ideal, itype,                  6053
    units, desc, is_con) in TREE:                                        6054
        ind = Indicator.query.filter_by(code=code).first()               6055
        if ind is None:                                                  6056
            ind = Indicator(code=code)                                   6057
            db.session.add(ind)                                          6058
        ind.name = name                                                  6059
        ind.level = level                                                6060
        ind.parent_code = parent                                         6061
        ind.unit = unit                                                  6062
        ind.ideal_value = ideal                                          6063
        ind.ideal_type = itype                                           6064
        ind.enabled_units = units                                        6065
        ind.description = desc                                           6066
        ind.is_constraint = is_con                                       6067
        ind.is_evaluative = (level in (2, 3))                            6068
        n += 1                                                           6069
    db.session.commit()                                                  6070
    return n                                                             6071
def ideal_type_label(code):                                              6072
    return {"benefit": "越大越优", "cost": "越小越优",                    6073
            "mid": "适中型", "discrete": "离散赋值"}.get(code,            6074
    code or "—")                                                         6075
# ==== app/templates/ahp/experts.html ====                               6076
{% extends "base.html" %}                                                6077
{% block title %}多专家判断矩阵{% endblock %}                             6078
{% block content %}                                                      6079
<div class="card">                                                       6080
  <h2>多专家判断矩阵 · 录入与汇总</h2>                                    6081
  <p class="muted">                                                      6082
    论文明确权重由「组织 １０ 位专家对各项指标权重进行打分」确定。        6083
    此页支持多位专家分别录入判断矩阵，再用几何平均汇总，                  6084
    保持矩阵互反性，并重新做一致性检验。                                  6085
  </p>                                                                   6086
  <form method="get" class="actions" style="margin-bottom:14             6087
    px;">                                                                6088
    <div>                                                                6089
      <label class="field__label" for="unit-type">微地质单元类型</l      6090
    abel>                                                                6091
      <select id="unit-type" name="unit_type"                            6092
    onchange="this.form.submit()">                                       6093
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}                   6094
        <option value="{{ t }}" {% if unit_type == t                     6095
    %}selected{% endif %}>{{ t }}</option>                               6096
        {% endfor %}                                                     6097
      </select>                                                          6098
    </div>                                                               6099
    <div>                                                                6100
```

```
      <label class="field__label" for="level">指标层级</label>
      <select id="level" name="level"
    onchange="this.form.submit()">
        <option value="1" {% if level == 1 %}selected{%
    endif %}>一级指标</option>
        <option value="2" {% if level == 2 %}selected{%
    endif %}>二级指标</option>
      </select>
    </div>
  </form>
  <p class="muted">
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif
    %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
  {% if rows %}
  <table>
    <caption class="muted">已录入 {{ rows|length }}
    位专家的矩阵</caption>
    <thead>
      <tr>
        <th scope="col">专家</th>
        <th scope="col">矩阵规模</th>
        <th scope="col">该矩阵 CR</th>
        <th scope="col">一致性</th>
        <th scope="col">录入时间</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td>{{ r.expert_name }}</td>
        <td class="num">{{ n }} × {{ n }}</td>
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
    not none else "—" }}</td>
        <td>
          {% if r.cr is not none and r.cr < 0.1 %}<span
    class="tag tag--ok">通过</span>
          {% else %}<span class="tag tag--warn">不通过</span>{%
    endif %}
        </td>
        <td class="mono nowrap">{{ r.created_at.strftime("%Y
    -%m-%d %H:%M") if r.created_at else "—" }}</td>
        <td>
          <form method="post" action="/ahp/experts/{{ r.id
    }}/delete"
                onsubmit="return confirm('确定删除专家「{{
```

```
    r.expert_name }}」的矩阵吗？');">                                    6151
            {{ form.hidden_tag() }}                                      6152
            <button type="submit" class="btn btn--sm                     6153
    btn--danger">删除</button>                                           6154
          </form>                                                        6155
        </td>                                                            6156
      </tr>                                                              6157
      {% endfor %}                                                       6158
    </tbody>                                                             6159
</table>                                                                 6160
<p class="actions" style="margin-bottom:10px;">                          6161
    <a class="btn" href="/ahp/experts/aggregate?unit_type={{             6162
    unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{%              6163
    endif %}">                                                           6164
      汇总计算（几何平均）                                               6165
    </a>                                                                 6166
</p>                                                                     6167
{% else %}                                                               6168
<p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>          6169
{% endif %}                                                              6170
<h3>录入一位专家的判断矩阵</h3>                                          6171
<form method="post">                                                     6172
    {{ form.hidden_tag() }}                                              6173
    <input type="hidden" name="action" value="add">                      6174
    <input type="hidden" name="unit_type" value="{{                      6175
    unit_type }}">                                                       6176
    <input type="hidden" name="level" value="{{ level }}">               6177
    {% if l1 %}<input type="hidden" name="l1" value="{{ l1               6178
    }}">{% endif %}                                                      6179
    <div class="field">                                                  6180
      <label class="field__label" for="expert-name">专家姓名</label>     6181
    <input id="expert-name" type="text" name="expert_name"               6182
      <input                                                             6183
        style="width:220px;" placeholder="如：张工">                     6184
    </div>                                                               6185
    <table style="width:auto;">                                          6186
      <thead>                                                            6187
        <tr>                                                             6188
          <th></th>                                                      6189
          {% for name in names %}<th scope="col">{{ name                 6190
    }}</th>{% endfor %}                                                  6191
        </tr>                                                            6192
      </thead>                                                           6193
      <tbody>                                                            6194
        {% for i in range(n) %}                                          6195
        <tr>                                                             6196
          <th scope="row">{{ names[i] }}</th>                            6197
          {% for j in range(n) %}                                        6198
            {% if i == j %}                                              6199
            <td class="muted num">1</td>                                 6200
```

```
                {% elif i < j %}                                                6201
                <td>                                                            6202
                  <label class="muted" style="position:absolute;                6203
    left:-9999px;"                                                              6204
                         for="e-{{ i }}-{{ j }}">{{ names[i] }}                 6205
    相对 {{ names[j] }} 的重要性</label>                                        6206
                  <input id="e-{{ i }}-{{ j }}" type="number"                   6207
    step="0.01" min="0.01"                                                      6208
                         name="m_{{ i }}_{{ j }}" value="{{                     6209
    default[i][j] }}">                                                          6210
                </td>                                                           6211
                {% else %}                                                      6212
                <td class="muted num">{{ default[i][j] }}</td>                  6213
                {% endif %}                                                     6214
              {% endfor %}                                                      6215
            </tr>                                                               6216
            {% endfor %}                                                        6217
          </tbody>                                                              6218
    </table>                                                                    6219
    <p class="actions" style="margin-top:12px;">                                6220
      <button type="submit" class="btn">录入该专家矩阵</button>                 6221
      <a class="btn btn--ghost" href="/ahp/indicators">指标维护<                6222
    /a>                                                                         6223
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>                   6224
    </p>                                                                        6225
  </form>                                                                       6226
</div>                                                                          6227
{% endblock %}                                                                  6228
# ==== app/templates/ahp/experts_aggregate.html ====                            6229
{% extends "base.html" %}                                                       6230
{% block title %}多专家矩阵汇总{% endblock %}                                   6231
{% block content %}                                                             6232
<div class="card">                                                              6233
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{%         6234
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                                 6235
  <p class="muted">                                                             6236
    对各位专家的判断矩阵按对应元素取几何平均：                                  6237
    a<sub>ij</sub> = (a<sub>ij</sub><sup>(1)</sup> × … ×                        6238
    a<sub>ij</sub><sup>(k)</sup>)<sup>1/k</sup>。                               6239
    该方式可保持矩阵的互反性，汇总后重新计算权重与一致性比例。                6240
  </p>                                                                          6241
  {% if not result %}                                                           6242
  <p class="muted">                                                             6243
    该范围内尚未录入专家矩阵。                                                  6244
    <a href="/ahp/experts?unit_type={{ unit_type                                6245
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif                         6246
    %}">去录入</a>                                                              6247
  </p>                                                                          6248
  {% else %}                                                                    6249
  <div class="stat-row">                                                        6250
```

```
    <div>                                                                6251
      <div class="stat__label">参与汇总的专家</div>                      6252
      <div class="stat__value">{{ result.count }} 位</div>               6253
    </div>                                                               6254
    <div>                                                                6255
      <div class="stat__label">λmax</div>                                6256
      <div class="stat__value mono">{{ result.lambda_max                 6257
    }}</div>                                                             6258
    </div>                                                               6259
    <div>                                                                6260
      <div class="stat__label">CI</div>                                  6261
      <div class="stat__value mono">{{                                   6262
    "%.4f"|format(result.ci) }}</div>                                    6263
    </div>                                                               6264
    <div>                                                                6265
      <div class="stat__label">CR（要求 &lt; 0.1）</div>                 6266
      <div class="stat__value mono">{{                                   6267
    "%.4f"|format(result.cr) }}</div>                                    6268
    </div>                                                               6269
    <div>                                                                6270
      <div class="stat__label">一致性检验</div>                          6271
      <div class="stat__value">                                          6272
        {% if result.consistent %}<span class="ok">通过</span>           6273
        {% else %}<span class="warn">不通过</span>{% endif %}            6274
      </div>                                                             6275
    </div>                                                               6276
  </div>                                                                 6277
  <h3>汇总后权重</h3>                                                    6278
  <table style="width:auto;">                                            6279
    <thead>                                                              6280
      <tr>                                                               6281
        <th scope="col">指标</th>                                        6282
        {% for name in names %}<th scope="col">{{ name                   6283
    }}</th>{% endfor %}                                                  6284
      </tr>                                                              6285
    </thead>                                                             6286
    <tbody>                                                              6287
      <tr>                                                               6288
        <th scope="row">权重</th>                                        6289
        {% for w in result.weights %}<td class="mono">{{ w               6290
    }}</td>{% endfor %}                                                  6291
      </tr>                                                              6292
    </tbody>                                                             6293
  </table>                                                               6294
  <h3>各专家矩阵的一致性</h3>                                            6295
  <table style="width:auto;">                                            6296
    <thead>                                                              6297
      <tr><th scope="col">专家</th><th scope="col">该矩阵                6298
    CR</th><th scope="col">判定</th></tr>                                6299
    </thead>                                                             6300
```

```
      <tbody>
        {% for r in rows %}
        <tr>
          <td>{{ r.expert_name }}</td>
          <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is
      not none else "—" }}</td>
          <td>
            {% if r.cr is not none and r.cr < 0.1 %}<span
      class="ok">通过</span>
            {% else %}<span class="warn">不通过</span>{% endif %}
          </td>
        </tr>
        {% endfor %}
      </tbody>
    </table>
    {% if not result.consistent and result.most_inconsistent
      %}
    <p style="padding:12px;border-radius:6px;background:var(--
      danger-soft);
                border-left:4px solid var(--danger);">
      <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{
      result.most_inconsistent.j + 1 }} 列的比较
      （当前 <span class="mono">{{ result.most_inconsistent.value
      |round(3) }}</span>，
      理想约 <span class="mono">{{ result.most_inconsistent.expec
      ted|round(3) }}</span>）。
      该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
    </p>
    {% endif %}
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/ahp/experts?unit_type={
      { unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1
      }}{% endif %}">返回录入</a>
      <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:680px;">
    <h2>编辑指标 · {{ ind.name }}</h2>
    <p class="muted">
      指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{
      ind.level }} 级，
      由论文指标体系确定，不可修改。
    </p>
```

```
<form method="post">
  {{ form.hidden_tag() }}
  <div class="field">
    <label class="field__label" for="name">指标名称</label>
    <input id="name" type="text" name="name"
style="width:100%;" required
           value="{{ ind.name }}">
  </div>
  <div class="field">
    <label class="field__label" for="unit">计量单位</label>
    <input id="unit" type="text" name="unit"
style="width:180px;"
           value="{{ ind.unit or '' }}">
  </div>
  <div class="field">
    <label class="field__label" for="ideal-type">标准化类型</la
bel>
    <select id="ideal-type" name="ideal_type">
      {% for code, label in [("benefit","越大越优"),("cost","越
小越优"),("mid","适中型"),("discrete","离散赋值")] %}
      <option value="{{ code }}" {% if ind.ideal_type ==
code %}selected{% endif %}>{{ label }}</option>
      {% endfor %}
    </select>
  </div>
  <div class="field">
    <label class="field__label" for="ideal-value">理想取值</la
bel>
    <input id="ideal-value" type="text"
name="ideal_value" style="width:100%;"
           value="{{ ind.ideal_value or '' }}">
    <p class="muted" style="margin:4px 0 0;">
      如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
    </p>
  </div>
  <fieldset style="border:1px solid
var(--line);border-radius:6px;padding:10px
14px;margin:0 0 12px;">
    <legend style="padding:0 6px;font-weight:600;">启用该指标的微
地质单元</legend>
    {% for t in unit_types %}
    <label class="checkbox" style="margin-right:18px;">
      <input type="checkbox" name="enabled_units"
value="{{ t }}"
             {% if t in ind.units_list() %}checked{%
endif %}>
      <span>{{ t }}</span>
    </label>
    {% endfor %}
    <p class="muted" style="margin:8px 0 0;">
```

```
            论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
          </p>
        </fieldset>
        <div class="field">
          <label class="checkbox" for="is-constraint">
            <input id="is-constraint" type="checkbox"
        name="is_constraint" value="1"
                   {% if ind.is_constraint %}checked{% endif %}>
            <span>同时作为限制性指标（参与一票否决判定）</span>
          </label>
        </div>
        <div class="field">
          <label class="field__label" for="description">取值依据说明</
        label>
          <textarea id="description" name="description"
        style="min-height:88px;">{{ ind.description or ''
        }}</textarea>
        </div>
        <p class="actions" style="margin-top:16px;">
          <button type="submit" class="btn">保存</button>
          <a class="btn btn--ghost" href="/ahp/indicators">返回指标列
        表</a>
        </p>
      </form>
</div>
{% endblock %}
# ==== app/templates/ahp/indicators.html ====
{% extends "base.html" %}
{% block title %}指标定义维护{% endblock %}
{% block content %}
<div class="card">
  <h2>指标体系维护</h2>
  <p class="muted">
    按论文表 2 与表 3 建立的指标树。此处可核对指标名称、单位、标准化类型、
    理想取值与取值依据，并按需调整启用该指标的微地质单元。
    计算时的权威取值仍以 <span class="mono">app/utils/constants.py</sp
    an> 为准，
    本页用于展示与核对。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/ahp/">返回权重计算</a>
    <a class="btn btn--sm btn--ghost"
    href="/ahp/experts">多专家矩阵</a>
    <form method="post" action="/ahp/indicators/reset"
    style="display:inline;"
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。
    确定继续？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
```

```
    btn--danger">按论文取值重置</button>
    </form>
</p>
{% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
    "三级指标")] %}
{% set level_items = all_ind|selectattr("level",
    "equalto", level)|list %}
{% if level_items %}
<h3>{{ title }}（{{ level_items|length }} 项）</h3>
<table>
    <thead>
        <tr>
            <th scope="col">指标</th>
            <th scope="col">上级指标</th>
            <th scope="col">单位</th>
            <th scope="col">标准化类型</th>
            <th scope="col">理想取值</th>
            <th scope="col">启用单元</th>
            <th scope="col">限制性</th>
            <th scope="col">操作</th>
        </tr>
    </thead>
    <tbody>
        {% for i in level_items %}
        <tr>
            <td style="text-align:left;">
                <span class="mono muted">{{ i.code
    }}</span><br>{{ i.name }}
            </td>
            <td>{{ (all_ind|selectattr("code", "equalto",
    i.parent_code)|list|first).name if i.parent_code else
    "—" }}</td>
            <td>{{ i.unit or "—" }}</td>
            <td>{{ type_label(i.ideal_type) }}</td>
            <td class="mono">{{ i.ideal_value or "—" }}</td>
            <td style="text-align:left;">{{ i.enabled_units or
    "—" }}</td>
            <td>
                {% if i.is_constraint %}<span class="tag
    tag--warn">是</span>
                {% else %}<span class="tag tag--muted">否</span>{%
    endif %}
            </td>
            <td><a href="/ahp/indicators/{{ i.code
    }}/edit">编辑</a></td>
        </tr>
        {% endfor %}
    </tbody>
</table>
{% endif %}
```

```
  {% endfor %}
</div>
{% endblock %}
# ==== app/templates/auth/edit_user.html ====
{% extends "base.html" %}
{% block title %}编辑用户 · {{ user.username }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
  <h2>编辑用户 · {{ user.username }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="role">角色</label>
      <select id="role" name="role">
        {% for code, label in role_labels.items() %}
        <option value="{{ code }}" {% if user.role == code
    %}selected{% endif %}>{{ label }}</option>
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="checkbox" for="active">
        <input id="active" type="checkbox" name="active"
    value="1"
               {% if user.active %}checked{% endif %}>
        <span>启用该账号</span>
      </label>
      <p class="muted" style="margin:6px 0 0;">
        停用后该账号无法登录，但其历史操作记录会保留。
        系统至少需保留一名启用状态的管理员。
      </p>
    </div>
    <div class="field">
      <label class="field__label" for="password">重置密码（留空则不修改
    ）</label>
      <input id="password" type="password" name="password"
             autocomplete="new-password"
    style="width:280px;"
             placeholder="留空表示保持原密码">
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      <a class="btn btn--ghost" href="/users">返回</a>
    </p>
  </form>
</div>
{% endblock %}
# ==== app/templates/auth/logs.html ====
{% extends "base.html" %}
{% block title %}操作日志{% endblock %}
```

```
{% block content %}
<div class="card">
  <h2>操作日志</h2>
  <p class="muted">
    系统自动记录所有写操作（新建、编辑、删除、导入、执行评价等），
    便于追溯数据变更。失败的操作也会记录并标注 HTTP 状态码。
  </p>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm btn--ghost"
    href="/users">返回用户管理</a>
  </p>
  {% if not records %}
  <p class="muted">暂无操作记录。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}
    / {{ pages }} 页</caption>
    <thead>
      <tr>
        <th scope="col">时间</th>
        <th scope="col">操作人</th>
        <th scope="col">操作</th>
        <th scope="col">对象</th>
        <th scope="col">说明</th>
      </tr>
    </thead>
    <tbody>
      {% for r in records %}
      <tr>
        <td class="mono nowrap">{{ r.created_at.strftime("%Y
    -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>
        <td>{{ users_map.get(r.user_id, "—") }}</td>
        <td style="text-align:left;">{{ r.action }}</td>
        <td>
          {% if r.target_type %}{{ r.target_type }}{% if
    r.target_id %}#{{ r.target_id }}{% endif %}
          {% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ r.detail or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% if pages > 1 %}
  <nav class="actions" aria-label="日志分页">
    {% if page > 1 %}
    <a class="btn btn--sm btn--ghost" href="/logs?page={{
    page - 1 }}">上一页</a>
    {% endif %}
```

```
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>
      {% if page < pages %}
      <a class="btn btn--sm btn--ghost" href="/logs?page={{
      page + 1 }}">下一页</a>
      {% endif %}
    </nav>
    {% endif %}
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/error.html ====
{% extends "base.html" %}
{% block title %}{{ code }} · {{ message }}{% endblock %}
{% block content %}
<div class="card" style="max-width:560px;">
    <h2>{{ code }} · {{ message }}</h2>
    <p class="muted">
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
    <h2>批量导入指标数据</h2>
    <p class="muted">
      支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
      系统按「单元类型」匹配图斑内已划分的单元后写入数据。
      先下载模板按格式填写，可减少出错。
    </p>
    <p class="actions" style="margin-bottom:14px;">
      <a class="btn btn--sm btn--ghost"
      href="/indicator/import/template">下载导入模板（CSV）</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">返回指标数据</a>
    </p>
    <form method="post" enctype="multipart/form-data">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="patch-id">目标矿山图斑</lab
        el>
        <select id="patch-id" name="patch_id" required>
          <option value="">— 请选择 —</option>
          {% for p in patches %}
          <option value="{{ p.id }}">{{ p.name }} ({{
```

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      p.units.count() }} 个单元）</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="datafile">数据文件</label
      >
        <input id="datafile" type="file" name="datafile"
               accept=".csv,.xlsx,.xls" required>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">开始导入</button>
      </p>
    </form>
    <h3>模板列说明</h3>
    <p class="muted">
      共 {{ headers|length }} 列。留空表示该指标「不适用」。
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
      （按论文注释视为未受污染）。
    </p>
    <p class="mono muted" style="line-height:1.9;">{{
      headers|join(" , ") }}</p>
    <h3>导入规则</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
```

```
  </div>
  {% if report.errors %}
  <h3>未导入的行</h3>
  <ul class="muted" style="padding-left:20px;line-height:1.9;">
    {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
  </ul>
  {% else %}
  <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
  </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/report/print.html ====
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0 6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom: 4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0 6px; }
  .sub { text-align: center; color: #6b7280; font-size: 9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin: 8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px; text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
```

```
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0 ·
    生成时间 {{ generated_at }}
  </p>
  <h2>一、图斑基本信息</h2>
  <table class="kv">
    <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
    <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
    }}</td></tr>
    <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
    patch.area_ha is not none else "—" }}</td></tr>
    <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—"
    }}</td></tr>
    <tr><th>开采方式</td><td>{{ patch.mining_method or "—"
    }}</td></tr>
    <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entit
    y_missing else "否" }}</td></tr>
    {% if reference %}
    <tr><th>参照生态系统</td>
        <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%，
            多样性指数基准 {{ reference.shannon_benchmark
    }}</td></tr>
    {% endif %}
  </table>
  <h2>二、评价结论</h2>
  <table class="kv">
    <tr><th>最终评价因子 F</td><td class="mono"><strong>{{
    "%.4f"|format(result.final_score) }}</strong></td></tr>
    <tr><th>适宜性等级</td><td><strong>{{ result.grade
    }}</strong></td></tr>
    <tr><th>判定依据</td><td>{{ result.grade_info.meaning
    }}</td></tr>
    <tr><th>推荐措施</td><td>{{ result.grade_info.measure
    }}</td></tr>
  </table>
  <h2>三、各微地质单元得分</h2>
  <table>
```

```
    <thead>
      <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
    F'</th><th>加权贡献</th></tr>
    </thead>
    <tbody>
      {% for u in result.unit_results %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.unit_score)
    }}</td>
        <td class="mono">{{ "%.4f"|format(u.weighted)
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <div class="chart">{{ chart_units|safe }}</div>
  <h2>四、一级指标权重分布</h2>
  <div class="chart">{{ chart_weight|safe }}</div>
  <h2>五、得分明细</h2>
  {% for u in result.unit_results %}
  <h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
    100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
  <table>
    <thead>
      <tr>
        <th>一级指标</th><th>二级指标</th><th>实测值</th>
        <th>标准化值</th><th>权重</th><th>贡献</th>
      </tr>
    </thead>
    <tbody>
      {% for d in u.details %}
      <tr>
        <td>{{ d.l1 }}</td>
        <td class="left">{{ d.l2 }}</td>
        <td class="mono left">{{ "—" if d.raw is none else
    d.raw }}</td>
        <td class="mono">{{ "—" if d.std_value is none else
    "%.4f"|format(d.std_value) }}</td>
        <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
        <td class="mono">{{ "—" if d.contribution is none
    else "%.4f"|format(d.contribution) }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  <p class="note">
    一级指标得分：
```

```
      {% for k, v in u.l1_scores.items() %}{{ k }} = {{ "%.4f"|format(v) }}{% if not loop.last %};{% endif %}{% endfor %}
   </p>
   {% endfor %}
   <h2>六、评价依据说明</h2>
   <ul class="note">
      <li>指标标准化按论文式（８） 执行，按指标性质分为越大越优、越小越优、适中型与离散赋值四类。</li>
      <li>单元评价因子按论文式（９） 由二级指标权重加权求得；整矿评价因子由各微地质单元按面积占比加权合成。</li>
      <li>适宜性等级按论文表 4 划分：F&lt;０．３ 不适宜、０．３≤F≤０．７ 基本适宜、F≥０．７ 适宜。</li>
      <li>指标权重取自论文表 3，按岩质边坡、碎石土边坡、平台三类微地质单元分别确定。</li>
   </ul>
   <p class="sig">
      评价人（签字）：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿ 日期：＿＿＿＿＿＿＿＿＿＿＿＿
   </p>
</body>
</html>
#  ==== app/utils/__init__.py ====
#  -*- coding: utf-8 -*-
#  ==== app/utils/browser.py ====
#  -*- coding: utf-8 -*-
"""HTML 转 PDF（M9 报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
      r"C:\Program Files\Google\Chrome\Application\chrome.exe",
      r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
      r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
      r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
#  允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
      """定位可用的浏览器可执行文件，找不到返回 None。"""
      override = os.environ.get(ENV_KEY)
      if override and os.path.isfile(override):
```

```
            return override                                                    6901
        for p in CANDIDATES:                                                   6902
            if os.path.isfile(p):                                              6903
                return p                                                       6904
        for name in ("chrome", "msedge", "chromium"):                          6905
            found = shutil.which(name)                                         6906
            if found:                                                          6907
                return found                                                   6908
        return None                                                            6909
def html_to_pdf(html, timeout=90):                                             6910
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""                6911
    exe = browser_path()                                                       6912
    if exe is None:                                                            6913
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"              6914
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)            6915
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")                                6916
    tag = uuid.uuid4().hex[:8]                                                 6917
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)                        6918
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)                          6919
    try:                                                                       6920
        with open(html_path, "w", encoding="utf-8") as f:                      6921
            f.write(html)                                                      6922
        cmd = [                                                                6923
            exe, "--headless=new", "--disable-gpu",                            6924
    "--no-sandbox",                                                            6925
            "--no-first-run", "--no-default-browser-check",                    6926
            "--no-pdf-header-footer",                                          6927
            "--virtual-time-budget=4000",                                      6928
            "--print-to-pdf=%s" % pdf_path,                                    6929
            "file:///" + html_path.replace("\\", "/"),                         6930
        ]                                                                      6931
        p = subprocess.run(cmd, capture_output=True,                           6932
    text=True,                                                                 6933
                           timeout=timeout,                                    6934
    encoding="utf-8", errors="replace")                                        6935
        if not os.path.isfile(pdf_path) or                                     6936
    os.path.getsize(pdf_path) < 800:                                           6937
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or                 6938
    "").strip()[:200]                                                          6939
        with open(pdf_path, "rb") as f:                                        6940
            return f.read(), None                                              6941
    except subprocess.TimeoutExpired:                                          6942
        return None, "生成 PDF 超时，请稍后重试"                                    6943
    except Exception as e:                                                     6944
        return None, "生成 PDF 失败：%s" % e                                       6945
    finally:                                                                   6946
        shutil.rmtree(tmpdir, ignore_errors=True)                              6947
# ==== app/utils/charts.py ====                                                6948
# -*- coding: utf-8 -*-                                                        6949
"""轻量图表（M9）。                                                             6950
```

```
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
        return str(v)
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,
    value_w=76):
    """水平条形图。items: [(标签, 数值, 显示文本)]"""
    if not items:
        return '<p class="muted">暂无数据</p>'
    n = len(items)
    height = n * (bar_h + gap) + 10
    max_v = max((abs(float(v)) for _, v, _ in items),
    default=1) or 1
    body_w = width - label_w - value_w
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"
    height="%d" '
             'role="img" aria-label="条形图" '
             'style="max-width:%dpx;font-family:inherit;">'
    % (width, height, height, width)]
    for i, (label, value, disp) in enumerate(items):
        y = i * (bar_h + gap) + 5
        frac = (abs(float(value)) / max_v) if max_v else 0
        w = max(2.0, body_w * frac)
        fill = GRAYS[i % len(GRAYS)]
        parts.append(
            '<text x="%d" y="%d" text-anchor="end"
    font-size="12" fill="%s">%s</text>'
            % (label_w - 10, y + bar_h * 0.72, INK,
    escape(str(label))))
        parts.append(
            '<rect x="%d" y="%d" width="%.1f" height="%d"
    rx="3" fill="%s"/>'
            % (label_w, y, w, bar_h, fill))
        parts.append(
            '<text x="%d" y="%d" font-size="12" fill="%s" '
```

```
            'style="font-variant-numeric:tabular-nums;">%s</
text>'
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,
escape(str(disp))))
    parts.append('</svg>')
    return "".join(parts)
def donut(items, size=210, thickness=34):
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。
    """
    vals = [(l, float(v), d) for l, v, d in items if
    float(v) > 0]
    if not vals:
        return '<p class="muted">暂无数据</p>'
    total = sum(v for _, v, _ in vals) or 1
    r = (size - thickness) / 2
    cx = cy = size / 2
    import math
    parts = ['<svg viewBox="0 0 %d %d" width="%d"
    height="%d" role="img" '
             'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
            % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
```

```
                '<th scope="col">项目</th><th                                7051
    scope="col">数值</th>'                                                   7052
                '<th scope="col">占比</th></tr></thead><tbody>']             7053
    for i, (label, v, disp) in enumerate(vals):                              7054
        legend.append(                                                       7055
            '<tr><td style="text-align:left;">'                              7056
            '<span style="display:inline-block;width:10px;he                 7057
    ight:10px;'                                                              7058
            'border-radius:2px;background:%s;margin-right:6p                 7059
    x;"></span>%s</td>'                                                      7060
            '<td class="mono">%s</td><td                                     7061
    class="mono">%.1f%%</td></tr>'                                           7062
            % (GRAYS[i % len(GRAYS)], escape(str(label)),                    7063
    escape(str(disp)),                                                       7064
               100.0 * v / total))                                           7065
    legend.append('</tbody></table>')                                        7066
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p                 7067
    x;align-items:center;">'                                                 7068
            + "".join(parts) + "".join(legend) + '</div>')                   7069
# ==== app/utils/schema.py ====                                              7070
# -*- coding: utf-8 -*-                                                      7071
"""轻量结构迁移。                                                             7072
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。                  7073
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，                   7074
使旧数据库无需重建即可用上新字段。                                           7075
"""                                                                          7076
from sqlalchemy import text                                                  7077
# （表名， 列名， 列定义）                                                   7078
REQUIRED_COLUMNS = [                                                         7079
    # M2 遥感影像挂接                                                        7080
    ("mine_patches", "image_path", "VARCHAR(256)"),                          7081
    ("mine_patches", "image_note", "VARCHAR(256)"),                          7082
    ("mine_patches", "image_uploaded_at", "DATETIME"),                       7083
    # M6 指标维护                                                            7084
    ("indicators", "enabled_units", "VARCHAR(128)"),                         7085
    ("indicators", "description", "TEXT"),                                   7086
    # M6 多专家矩阵                                                          7087
    ("expert_matrices", "unit_type", "VARCHAR(32)"),                         7088
    ("expert_matrices", "l1_name", "VARCHAR(64)"),                           7089
    ("expert_matrices", "cr", "FLOAT"),                                      7090
]                                                                            7091
def ensure_schema(db):                                                       7092
    """为已有表补齐缺失列。返回补列清单。"""                                 7093
    added = []                                                               7094
    try:                                                                     7095
        with db.engine.connect() as conn:                                    7096
            for table, column, ddl in REQUIRED_COLUMNS:                      7097
                rows = conn.execute(text("PRAGMA                             7098
    table_info(%s)" % table)).fetchall()                                     7099
                if not rows:                                                 7100
```

```
                    continue                                    #          7101
    表还不存在，create_all 会建                                              7102
                existing = {r[1] for r in rows}                             7103
                if column in existing:                                      7104
                    continue                                                7105
                conn.execute(text(                                          7106
                    "ALTER TABLE %s ADD COLUMN %s %s" %                     7107
    (table, column, ddl)))                                                  7108
                conn.commit()                                               7109
                added.append("%s.%s" % (table, column))                     7110
    except Exception as e:                                      #          7111
    迁移失败不应阻断启动                                                     7112
        print("[schema] 结构检查跳过：%s" % e)                               7113
    return added                                                            7114
# ==== app/utils/uploads.py ====                                           7115
# -*- coding: utf-8 -*-                                                    7116
"""上传文件处理（M2 遥感影像、M7 批量导入）。"""                                 7117
import os                                                                   7118
import re                                                                   7119
import uuid                                                                 7120
from datetime import datetime                                               7121
from werkzeug.utils import secure_filename                                  7122
def _safe_ext(filename):                                                    7123
    return os.path.splitext(filename)[1].lower().lstrip(".")                7124
def ext_allowed(filename, allowed):                                         7125
    return _safe_ext(filename) in allowed                                   7126
def save_upload(file_storage, subdir, allowed_ext,                          7127
    upload_root):                                                           7128
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。      7129
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。                    7130
    """                                                                     7131
    if file_storage is None or not file_storage.filename:                   7132
        return None, "未选择文件"                                            7133
    ext = _safe_ext(file_storage.filename)                                  7134
    if ext not in allowed_ext:                                              7135
        return None, "不支持的文件类型：.%s" % ext                           7136
    target_dir = os.path.join(upload_root, subdir)                          7137
    os.makedirs(target_dir, exist_ok=True)                                  7138
    # 仅保留安全字符，并加时间戳与随机串防重名                                    7139
    base = secure_filename(os.path.splitext(file_storage.fil                7140
    ename)[0]) or "file"                                                    7141
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or                     7142
    "file"                                                                  7143
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M                7144
    %S"),                                                                   7145
                         uuid.uuid4().hex[:6], ext)                         7146
    abs_path = os.path.join(target_dir, name)                               7147
    file_storage.save(abs_path)                                             7148
    rel = os.path.join("uploads", subdir,                                   7149
    name).replace("\\", "/")                                                7150
```

```
    return rel, None                                                    7151
def remove_upload(rel_path, upload_root):                               7152
    """删除已保存的上传文件（相对 static 的路径）。"""                    7153
    if not rel_path:                                                    7154
        return                                                          7155
    abs_path = os.path.join(upload_root,                                7156
    rel_path.replace("uploads/", "", 1))                                7157
    try:                                                                7158
        if os.path.isfile(abs_path):                                    7159
            os.remove(abs_path)                                         7160
    except OSError:                                                     7161
        pass                                                            7162
# ==== app/views/__init__.py ====                                       7163
# -*- coding: utf-8 -*-                                                 7164
```